

QUADCOPTER DELIVERY SIMULATION

A Minor Specialization Project Report

*Submitted in Partial Fulfillment of the
Requirements for the Minor Degree Specialization of*

Robotics

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CERTIFICATE OF ORIGINALITY OF WORK

I hereby declare that the project work entitled "**Quadcopter Delivery Simulation**" submitted in partial fulfillment of the requirements for the minor degree specialization in Robotics, Department of Mechanical Engineering, School of Technology, Pandit Deendayal Energy University, Gandhinagar, is an authentic record of work carried out under the supervision of **Dr. Ravi Kant**.

I further affirm that this report has been prepared in accordance with the university's academic integrity standards, that all sources consulted have been duly cited, and that no portion of this work has been previously submitted to any institution for the award of any degree or academic credential.

I acknowledge that any violation of academic honesty norms will render me liable to disciplinary proceedings as per university regulations.

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CERTIFICATE FROM THE PROJECT SUPERVISOR

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In my assessment, the quality and depth of the work meets the standards required for acceptance by the examination committee. The findings and results presented herein have not, to the best of my knowledge, been submitted to any other academic institution for the purpose of obtaining a degree or diploma.

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Any project of meaningful depth is rarely the product of one person working in isolation. As I look back on the journey that led to the completion of this capstone work on autonomous quadcopter delivery simulation, I find myself genuinely indebted to a wide circle of people whose contributions, in ways both visible and invisible, made this possible.

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project rests. Their work is cited throughout these pages and represents the cumulative wisdom that made it possible to attempt something this technically ambitious within a semester-length project.

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ABSTRACT

Unmanned aerial vehicles of the multirotor category, and quadcopters in particular, have attracted intense academic and industrial attention owing to their mechanical simplicity, ability to take off and land vertically, and capacity for precise stationary hover. These properties make them attractive candidates for autonomous last-mile delivery, aerial inspection, search-and-rescue operations, and precision agriculture. Despite their apparent mechanical simplicity, quadcopters present non-trivial control challenges: the vehicle is a fundamentally underactuated nonlinear system in which six spatial degrees of freedom must be managed through only four independent thrust inputs. This coupling between attitude and position, combined with the sensitivity of the system to external disturbances and parameter variations, makes robust autonomous flight a demanding engineering problem.

This project addresses that challenge through a rigorous simulation-based approach, developing and validating a complete autonomous quadcopter delivery system within the MATLAB Simulink and Simscape Multibody computational environment. A geometrically and physically representative six-degree-of-freedom rigid body model was constructed, incorporating the quadcopter frame, motor mounts, and propeller assemblies as interconnected rigid bodies governed by Newton-Euler dynamics. Actuator subsystems were modeled with first-order motor dynamics, aerodynamically derived thrust and reaction torque relationships, and a motor mixer that translates aggregate control demands into per-motor speed commands. Sensor models for an inertial measurement unit and GPS receiver were included to complete the closed-loop feedback chain.

The flight control architecture is organized as a hierarchy of nested PID loops. An outer position loop, operating at 50 Hz, computes desired attitude commands from position and velocity tracking errors. An inner attitude loop, operating at 500 Hz, stabilizes roll, pitch, and yaw using high-bandwidth PID controllers with IMU feedback. A gravity-compensating feedforward term in the altitude channel ensures that the integral action addresses only residual disturbances. All controller gains were systematically tuned using a combination of theoretical frequency-domain guidelines and iterative simulation-based refinement.

Smooth reference trajectories are generated by a polynomial interpolation engine that fits fifth-degree functions across each flight segment, guaranteeing continuity of position, velocity, and acceleration at waypoint boundaries. The delivery mission covers eight defined waypoints spanning takeoff, horizontal transit, descent to drop altitude, package release, ascent, and return

landing. Payload release is achieved through a position-triggered constraint switch in Simscape Multibody that disables the weld joint between the package and the drone body, after which the package undergoes physically realistic free-fall dynamics.

Simulation results across a 35-second mission demonstrate accurate trajectory following with negligible steady-state position error and well-damped transient behavior. Attitude angles remained stable and closely tracked their commanded references throughout. All four motor thrusts maintained balanced load sharing between 2.8 N and 3.5 N. Package release was triggered precisely at the designated spatial coordinates, and the subsequent free-fall trajectory matched theoretical ballistic predictions. The system also demonstrated robustness to the sudden mass reduction caused by package release, recovering smoothly through integral action in the altitude controller.

Keywords: *Quadcopter, Autonomous Delivery, UAV, 6-DOF Dynamics, PID Control, Simscape Multibody, Trajectory Generation, Motor Mixer, Payload Release, MATLAB Simulink*

TABLE OF CONTENTS

CERTIFICATE OF ORIGINALITY OF WORK.....	ii
CERTIFICATE FROM THE PROJECT SUPERVISOR.....	iii
ACKNOWLEDGEMENT	iv
ABSTRACT.....	vi
TABLE OF CONTENTS.....	viii
LIST OF FIGURES	xi
LIST OF TABLES.....	xii
NOMENCLATURE	xiii
Latin Symbols	xiii
Greek Symbols	xiv
Abbreviations	xiv
CHAPTER 1: INTRODUCTION.....	1
1.1 Background and Motivation.....	1
1.2 Scope of the Project.....	2
1.3 Significance of Simulation-Based Development	3
1.4 Organization of the Report.....	3
CHAPTER 2: LITERATURE REVIEW	5
2.1 Historical Development of Unmanned Aerial Vehicles.....	5
2.2 Quadcopter Modeling Approaches.....	5
2.3 Control Strategies for Multicopter Systems	6
2.4 Autonomous Delivery Systems in Literature and Practice	7
2.5 Gaps Identified in Existing Literature.....	8
CHAPTER 3: PROBLEM DEFINITION AND OBJECTIVES	9
3.1 Problem Statement	9
3.2 Project Objectives	9
CHAPTER 4: QUADCOPTER DYNAMICS.....	11
4.1 Configuration and Degrees of Freedom.....	11
4.2 Underactuation and Its Control Implications	12
4.3 Forces and Moments	12
4.4 Motor Dynamics.....	14
4.5 Gyroscopic Effects	14
CHAPTER 5: METHODOLOGY	15
5.1 Overview of the Development Approach.....	15
5.2 Phase 1 — Mechanical and Fundamental System Modeling.....	15

5.3 Phase 2 — Actuator and Dynamic Subsystems	15
5.4 Phase 3 — Control and Navigation Systems	16
5.5 Phase 4 — Payload and Simulation Execution	16
CHAPTER 6: HARDWARE AND SOFTWARE ENVIRONMENT.....	19
6.1 Software Tools	19
6.2 Development Hardware Specifications.....	19
6.3 Conceptual Hardware Requirements for Real-World Deployment	20
6.4 Simulation Fidelity Considerations.....	21
CHAPTER 7: SYSTEM ARCHITECTURE	23
7.1 Overview	23
7.2 Subsystem Descriptions	23
CHAPTER 8: SUBSYSTEM EXPLANATION	25
8.1 Six-Degree-of-Freedom Dynamics Block.....	25
8.2 Trajectory Generator	25
8.3 Position Controller (Outer Loop)	26
8.4 Attitude Controller (Inner Loop).....	26
8.5 Motor Mixer	27
8.6 Propeller Thrust and Torque Modeling.....	28
8.7 Package Release Mechanism	28
CHAPTER 9: MATHEMATICAL MODELING	31
9.1 Coordinate Frames	31
9.2 Rotation Matrix	31
9.3 Translational Dynamics.....	32
9.4 Rotational Dynamics	32
9.5 Euler Angle Kinematics	33
9.6 Physical Parameters.....	34
CHAPTER 10: CONTROLLER DESIGN	35
10.1 Control Architecture Overview.....	35
10.2 Position PID Controllers	35
10.3 Attitude PID Controllers	35
10.4 PID Gain Tuning Methodology.....	36
10.5 Anti-Windup and Saturation Handling	37
CHAPTER 11: TRAJECTORY GENERATION.....	39
11.1 Overview and Motivation.....	39
11.2 Polynomial Trajectory Planning.....	39
11.3 Time Scaling and Segment Duration Selection.....	40
11.4 Defined Waypoints.....	41

CHAPTER 12: RESULTS AND DISCUSSION.....	43
12.1 Position Tracking Performance.....	43
12.2 Attitude Response	44
12.3 Motor Thrust Analysis	45
12.4 Propeller Speed Analysis	46
12.5 Package Drop Simulation.....	47
12.6 Overall System Stability Assessment.....	48
CHAPTER 13: SIMULATION ENVIRONMENT VALIDATION	49
13.1 Purpose and Approach	49
13.2 Hover Equilibrium Validation.....	49
13.3 Attitude Step Response Validation	50
13.4 Package Free-Fall Validation	51
CHAPTER 14: LIMITATIONS	53
14.1 Current Limitations of the Simulation	53
14.1.1 Absence of Wind and Atmospheric Disturbance Modeling.....	53
14.1.2 Ideal Sensor Models	53
14.1.3 No Obstacle Avoidance Capability	53
14.1.4 Simplified Aerodynamic Model	54
14.1.5 Battery and Power System Dynamics	54
14.1.6 Rigid Body Assumption for Payload.....	54
14.2 Summary of Limitations and Mitigations	55
CHAPTER 15: CONCLUSION	57
REFERENCES	59
APPENDIX A: SIMULATION PARAMETERS SUMMARY	61
APPENDIX B: MOTOR MIXER ALLOCATION MATRIX	63
APPENDIX C: TRAJECTORY COEFFICIENT DERIVATION	64

LIST OF FIGURES

Figure 4.1: Motor and Propeller Subsystem in Simscape Multibody

Figure 5.1: Top-Level Simulink Model showing Trajectory Generation, Control, and Simscape Multibody Plant

Figure 7.1: System Signal Flow Diagram — Position, Attitude, and Propeller Subsystems

Figure 8.1: Position and Attitude PID Controller Subsystem in Simulink

Figure 8.2: Simscape Multibody Quadcopter Assembly with Package Release Trigger

Figure 12.1: Position Tracking — Actual vs. Desired Trajectories for X, Y, Z Axes

Figure 12.2: Attitude Response — Pitch, Roll, and Yaw Angles vs. Desired References

Figure 12.3: Propeller Thrust for All Four Motors Throughout Mission Duration

Figure 12.4: Propeller Speed — Counter-Rotating Motor Pairs Throughout Mission

Figure 12.5: Package Location Displacement from Drop-Off Point After Release

Figure 13.1: Hover Equilibrium Validation — Thrust vs. Weight

Figure 13.2: Step Response Validation for Attitude Controller

LIST OF TABLES

Table 6.1: Software Tools Used in the Simulation Environment

Table 6.2: Conceptual Hardware Components for a Real-World Deployment

Table 9.1: Physical Parameters of the Quadcopter Simulation Model

Table 10.1: Tuned PID Gains for Position and Attitude Controllers

Table 11.1: Mission Waypoints for the Quadcopter Delivery Simulation

Table 13.1: Hover Equilibrium Validation Results

Table 13.2: Attitude Step Response Validation Metrics

Table 14.1: Identified Limitations and Proposed Mitigations

Table A.1: Complete Simulation Parameter Set

Table B.1: Motor Mixer Allocation Matrix

NOMENCLATURE

Latin Symbols

Symbol	Description	Unit
m	Total mass of quadcopter system (drone + payload)	kg
g	Standard gravitational acceleration (9.81)	m/s ²
L	Arm length: distance from vehicle centre to propeller hub	m
U_1	Total collective thrust command	N
U_2	Roll torque command	N·m
U_3	Pitch torque command	N·m
U_4	Yaw torque command	N·m
T_i	Individual thrust generated by propeller i ($i = 1-4$)	N
Q_i	Reaction torque from propeller i	N·m
K	Ratio of torque coefficient to thrust coefficient	–
R	3×3 rotation (Direction Cosine) matrix	–
I	Moment of inertia tensor (3×3, diagonal)	kg·m ²
p, q, r	Body-frame angular velocity components (roll, pitch, yaw rates)	rad/s
x, y, z	Inertial-frame position coordinates	m
C_t	Non-dimensional thrust coefficient	–
C_ϕ	Non-dimensional torque coefficient	–
A	Propeller disk area (πR^2)	m ²

K_p, K_i, K_d	PID proportional, integral, and derivative gains	varies
$p(t)$	Time-parameterized trajectory position function	m
$D(v)$	Aerodynamic drag force as a function of velocity	N

Greek Symbols

Symbol	Description	Unit
ϕ	Roll angle (rotation about x-axis)	rad / deg
θ	Pitch angle (rotation about y-axis)	rad / deg
ψ	Yaw angle (rotation about z-axis)	rad / deg
ω_i	Angular velocity of motor i	rad/s
ρ	Ambient air density (sea level: 1.225)	kg/m ³
ε	Positional tolerance for package release trigger	m
$\tau_\phi, \tau_\theta, \tau_\psi$	Roll, pitch, and yaw torque commands	N·m

Abbreviations

Abbreviation	Expansion
UAV	Unmanned Aerial Vehicle
VTOL	Vertical Takeoff and Landing
6-DOF	Six Degrees of Freedom
PID	Proportional-Integral-Derivative
LQR	Linear Quadratic Regulator
MPC	Model Predictive Control
SMC	Sliding Mode Control

IMU	Inertial Measurement Unit
GPS	Global Positioning System
DCM	Direction Cosine Matrix
ESC	Electronic Speed Controller
BLDC	Brushless Direct-Current (motor)
NED	North-East-Down coordinate convention
ENU	East-North-Up coordinate convention
RPM	Revolutions Per Minute
PDEU	Pandit Deendayal Energy University

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

The global logistics sector is in the midst of a structural transformation, driven by the explosive growth of e-commerce, the increasing urgency of medical supply chains, and consumer expectations for rapid, on-demand delivery. These pressures have converged to create intense interest in autonomous aerial delivery as a viable operational model. Among the various unmanned aerial platforms that have emerged to address these needs, the quadcopter has become the dominant choice for civilian autonomous applications.

The quadcopter's appeal stems from a combination of properties that are difficult to achieve simultaneously in any other configuration. It takes off and lands vertically, requiring no dedicated runway infrastructure. It can hover stationary with exceptional spatial precision, a capability that is critical for accurate payload delivery. Its mechanical architecture — four fixed-pitch propellers driven by independent brushless motors, arranged symmetrically about a central chassis — involves no swashplate, collective pitch linkage, or tail rotor, making it far less mechanically complex than conventional rotorcraft. And advances in lithium polymer battery technology, microcontroller capability, and MEMS-based sensors have brought the manufacturing cost of functional quadcopter platforms down dramatically over the past decade.

These characteristics have not gone unnoticed by industry. Several major technology and logistics companies have progressed well beyond initial demonstrations into active operational deployments. Amazon's Prime Air program has evolved through multiple generations of drone design and is conducting commercial deliveries in select markets. Google's Wing subsidiary has obtained regulatory approval for commercial operations in Australia and the United States. Zipline has built an extensive fixed-wing drone network for medical supply delivery in Rwanda, Ghana, and other countries, completing hundreds of thousands of commercial deliveries. These are not speculative futures — they are existing engineering accomplishments that set a clear context for the technical work in this project.

However, the gap between the appealing conceptual properties of a quadcopter and a reliable, deployable autonomous delivery system is substantial. The fundamental challenge lies in the vehicle's dynamics. A quadcopter operates in six-dimensional space — three translational and three rotational degrees of freedom — but possesses only four independent control inputs, the

four propeller thrust forces. This disparity defines it formally as an underactuated mechanical system. The consequences are far-reaching: horizontal motion cannot be achieved directly, but only by tilting the entire airframe so that a component of the thrust vector points in the desired direction. This means attitude and position are inextricably coupled, and the control system must manage this coupling without amplifying disturbances or driving the vehicle into instability.

The dynamics are additionally nonlinear and sensitive to parameter uncertainties, sensor noise, and atmospheric disturbances such as wind gusts. Translating the desired flight behavior into stable motor commands requires a carefully designed, well-tuned control architecture. And validating that architecture in hardware, without first understanding its behavior in simulation, carries significant risks of crash and damage.

This project was motivated by the need to develop a complete, physically faithful simulation environment within which the full design and validation cycle for an autonomous quadcopter delivery system can be conducted safely and efficiently. The MATLAB Simulink and Simscape Multibody platforms offer the necessary combination of high-fidelity physics modeling, control system design tools, and 3D visualization to make this possible.

1.2 Scope of the Project

The scope of this work encompasses the end-to-end design and simulation of an autonomous quadcopter delivery system. This includes the construction of a physically representative six-degree-of-freedom rigid body model in Simscape Multibody, the design and tuning of a hierarchical PID control system, the implementation of a polynomial trajectory generator, and the simulation of a complete delivery mission including autonomous package release and return to base. The entire project is simulation-based; no physical hardware was fabricated.

The simulation captures all dynamically significant phenomena: aerodynamic thrust and reaction torque from each propeller, first-order motor dynamics, body-frame-to-inertial-frame thrust transformation, gyroscopic effects from spinning propellers, aerodynamic drag on both the drone and the released package, and the abrupt mass change that occurs at payload release. The combined effect is a simulation environment that closely mirrors what a real deployment would encounter.

1.3 Significance of Simulation-Based Development

In modern engineering practice, simulation is not a substitute for hardware testing — it is a prerequisite for it. A simulation environment allows design parameters to be swept rapidly, failure modes to be identified without risk, and controller gains to be optimized before a single physical component is sourced. For a system as complex and dynamically sensitive as an autonomous aerial delivery platform, this pre-validation step is essential.

The specific value of MATLAB Simulink and Simscape Multibody in this context is the level of physical fidelity they support. Simscape Multibody models are not abstract block diagrams — they are assemblies of rigid bodies with defined mass properties, connected by joints with defined constraints, subject to forces and torques that are computed from first principles by the solver at each time step. The resulting behavior emerges from the physics of the model, not from manually specified equations, which means that errors in understanding the system will reveal themselves in the simulation, rather than only in hardware.

This approach also provides a clean path from simulation to hardware implementation. The controller architecture designed and validated in Simulink maps directly onto the software architecture of a real flight controller, and the tuned PID gains from simulation serve as a starting point for the hardware tuning process, dramatically reducing the iteration time.

1.4 Organization of the Report

Chapter 2 surveys the relevant published literature on UAV development, quadcopter modeling approaches, and flight control strategies. Chapter 3 formally defines the technical problem and articulates the specific objectives that guided the project. Chapter 4 provides a rigorous treatment of quadcopter flight dynamics. Chapter 5 describes the four-phase development methodology. Chapter 6 documents the hardware and software environment. Chapter 7 describes the overall system architecture and signal flow. Chapter 8 provides detailed explanations of each subsystem. Chapter 9 presents the complete mathematical modeling framework. Chapter 10 covers the controller design and gain tuning methodology. Chapter 11 explains the trajectory generation strategy. Chapter 12 presents and interprets the simulation results. Chapter 13 validates the simulation against analytical benchmarks. Chapter 14 discusses limitations and future directions. Chapter 15 concludes the work.

CHAPTER 2: LITERATURE REVIEW

2.1 Historical Development of Unmanned Aerial Vehicles

The engineering lineage of unmanned aerial vehicles is considerably longer than popular accounts suggest. Early radio-controlled aircraft were investigated as aerial weapons during and after the First World War, with the Kettering Bug aerial torpedo representing one of the earliest functioning autonomous aircraft designs. Military interest sustained research through the mid-twentieth century, producing surveillance and target drone programs that proliferated during the Vietnam War era. The Global Hawk, Predator, and Reaper programs that came to define modern military UAV doctrine were the culmination of this decades-long investment.

The civilian multirotor revolution is, by contrast, a product of the early twenty-first century. Three enabling technologies converged to make it possible: microprocessors small enough to fit on a palm-sized flight controller board became sufficiently powerful to run real-time attitude estimation and control loops at kilohertz rates; lithium polymer battery chemistry made energy densities available that could sustain useful flight durations in lightweight configurations; and brushless DC motors with electronic speed controllers offered the combination of high power-to-weight ratio and fast response needed for motor-based flight control. By 2010, consumer quadcopters had become widely available, and an explosion of academic research rapidly followed, producing the rich literature on which this project builds.

2.2 Quadcopter Modeling Approaches

The mathematical modeling of quadcopter flight dynamics has been approached through several complementary frameworks, each with distinct strengths. The Newton-Euler approach applies Newton's second law directly to the translational dynamics of the center of mass, and Euler's rotational equations to the angular dynamics about the center of mass. This method is favored in practice for its physical transparency and computational efficiency, making it well suited to real-time implementation. Luukkonen (2011) produced a widely cited treatment of quadcopter modeling using this framework, deriving the full nonlinear equations of motion in a form suitable for control system design.

The Lagrangian approach derives the equations of motion from the system's kinetic and potential energy through the Euler-Lagrange formalism. While this method is mathematically elegant and well suited to revealing structural symmetries in the dynamics, it tends to produce more complex

symbolic expressions and is less commonly implemented in real-time control applications. For systems with holonomic constraints, however, it can provide cleaner derivations than the Newton-Euler alternative.

A third approach, and the one adopted in this project, uses multibody simulation environments such as Simscape Multibody to represent the vehicle as an assembly of rigid bodies with defined physical properties, allowing the simulation engine to automatically derive and numerically integrate the equations of motion. This approach requires no manual derivation of the equations of motion and easily accommodates complex geometries, but provides less analytical transparency. It is particularly well suited to projects where physical accuracy and rapid prototyping are prioritized over symbolic tractability.

Bouabdallah's doctoral work at EPFL (2004) established many of the canonical results on quadcopter modeling and control, providing a rigorous treatment of the 6-DOF dynamics and a comparative evaluation of PID and LQR controllers. This work has been extensively cited in subsequent literature and influenced the choice of control architecture in this project.

2.3 Control Strategies for Multirotor Systems

PID control, arranged in a cascaded hierarchy, remains the dominant paradigm for both research and commercial flight control systems. Its success owes to its conceptual simplicity, moderate sensitivity to model uncertainty, and straightforward gain tuning procedures. Practically every commercial flight controller, from the open-source Betaflight and ArduPilot platforms to the proprietary systems in commercial delivery drones, employs some variant of cascaded PID. Hoffmann et al. (2007) provided an experimental validation of quadcopter PID control that demonstrated the achievable levels of tracking accuracy and disturbance rejection in near-real-world conditions.

Linear Quadratic Regulator control offers formal optimality guarantees and structured gain computation through the solution of the algebraic Riccati equation, but requires an accurate linearized model and provides no explicit robustness guarantees against nonlinear effects or large deviations from the operating point. Several researchers have demonstrated LQR-based quadcopter controllers with improved steady-state performance compared to manually tuned PID, but the dependence on model accuracy limits its appeal for deployment on vehicles with significant parameter uncertainty.

Model Predictive Control has attracted considerable research interest due to its ability to explicitly account for actuator saturation and state constraints during trajectory optimization. Mueller and D’Andrea (2013) demonstrated MPC-based aggressive trajectory interception on quadcopter platforms at ETH Zurich, achieving maneuvers impossible with conventional PID. The computational burden of solving a receding-horizon optimization problem at flight control rates remains a practical barrier, though embedded implementations on modern microcontrollers have become increasingly feasible.

Sliding Mode Control offers strong robustness guarantees in the presence of bounded disturbances and parameter variations by driving the system state onto a designated manifold in state space. Its main practical limitation is the chattering phenomenon, which arises from high-frequency switching and can excite structural resonances or wear actuators. Boundary-layer smoothing and higher-order sliding mode variants have been proposed to address this limitation. Adaptive and learning-based approaches, including neural network controllers and reinforcement learning policies, are active research directions but remain rare in deployed systems due to interpretability and certification challenges.

2.4 Autonomous Delivery Systems in Literature and Practice

Academic interest in drone-based delivery predates the current wave of commercial programs by a decade or more. Early work focused primarily on trajectory optimization for fixed-wing platforms. Research interest in multirotor delivery grew substantially following the work of Mellinger and Kumar (2011), who derived minimum-snap trajectory generation algorithms capable of producing physically realizable flight paths between arbitrary waypoints while minimizing the jerk — and hence the structural loading and control effort — along the trajectory. This work established the theoretical foundation for polynomial trajectory generation methods that are now standard in autonomous multirotor research.

On the commercial side, Amazon’s Prime Air program has iterated through multiple generations of delivery drones, from early octocopter configurations to the hybrid fixed-wing/multirotor MK30 design capable of deliveries within 60 minutes. Google’s Wing program operates a tether-lowering delivery mechanism that deposits packages on the ground without the drone needing to land, addressing the challenge of safe landing in cluttered urban environments. Zipline has built the world’s largest drone delivery network for medical supplies, having completed more than a million commercial deliveries across Rwanda, Ghana, and the United States using fixed-wing VTOL platforms.

Academic studies on payload delivery drones consistently highlight several recurring technical challenges: maintaining trajectory accuracy in the presence of wind disturbances, efficient flight path planning to maximize battery utilization, precise payload positioning at the designated drop location, and safe operation in airspace shared with manned aviation. These challenges frame the problem addressed by this project.

2.5 Gaps Identified in Existing Literature

While the individual components of autonomous quadcopter delivery systems — dynamics modeling, PID control, trajectory generation, and payload release — have each been studied extensively in isolation, comprehensive integrated simulation frameworks that address the full delivery mission end-to-end within a physically faithful multibody environment are less common in the academic literature. Most published simulation studies use simplified mathematical models that abstract away motor dynamics, propeller aerodynamics, and payload physics. This project contributes a fully integrated Simscape Multibody simulation that captures all of these phenomena simultaneously, providing a more realistic baseline for future hardware implementation.

CHAPTER 3: PROBLEM DEFINITION AND OBJECTIVES

3.1 Problem Statement

The central engineering challenge of this project is to design and validate, entirely through simulation, a fully autonomous quadcopter delivery system that accurately captures the coupled nonlinear physics of aerial flight while demonstrating reliable, end-to-end mission execution. The technical difficulty arises from several compounding factors that must be addressed simultaneously.

First, the quadcopter is a physically underactuated system: six spatial degrees of freedom must be controlled using only four independent actuator inputs, necessitating a hierarchical control architecture that manages the inherent coupling between translational and rotational dynamics. Second, those dynamics are inherently nonlinear and strongly sensitive to the operating condition, meaning that a controller designed at one operating point may perform poorly at another. Third, the delivery mission imposes a sequence of coupled objectives — stable hover, trajectory following, descent, payload release at a precise location, ascent, and return — that must all be satisfied by the same control system without explicit reprogramming between phases. Fourth, the package release event constitutes an abrupt step disturbance in the gravity-compensation equilibrium of the vehicle, which the control system must absorb smoothly to prevent altitude loss or instability.

3.2 Project Objectives

The following specific and measurable objectives guided the development of this project:

- Develop a physically accurate 6-DOF dynamic model of the quadcopter within Simscape Multibody, incorporating rigid body geometry, mass and inertia properties, motor dynamics, propeller aerodynamics, and aerodynamic drag.
- Implement a hierarchical PID control architecture with an outer position loop computing desired attitude references from position error, and an inner attitude loop stabilizing roll, pitch, and yaw at high bandwidth.
- Design and implement a fifth-degree polynomial trajectory generator that produces smooth, time-parameterized reference trajectories between user-defined waypoints, with continuity of position, velocity, and acceleration at all waypoint boundaries.

- Accurately model the conversion from motor angular velocity to aerodynamic thrust and reaction torque using the standard quadratic thrust and torque relationships.
- Design and simulate an autonomous package release mechanism that triggers precisely when the drone reaches the designated delivery coordinates within a defined spatial tolerance.
- Simulate the post-release free-fall trajectory of the package under the combined action of gravity and aerodynamic drag.
- Visualize the complete mission in three dimensions using the Simscape Multibody Mechanics Explorer.
- Log and analyze all critical performance metrics, including three-axis position error, three-axis attitude response, individual motor thrust outputs, and propeller speed profiles.
- Validate the simulation against analytical benchmarks including hover equilibrium and attitude step response predictions.

CHAPTER 4: QUADCOPTER DYNAMICS

4.1 Configuration and Degrees of Freedom

The quadcopter belongs to the broader class of multirotor rotorcraft and is characterized by four fixed-pitch propellers, each driven by an independent brushless DC motor, mounted at the extremities of two crossing structural arms in an X or + arrangement. Unlike a conventional helicopter, in which thrust magnitude and direction are modulated by adjusting rotor blade pitch angles through a mechanical swashplate, the quadcopter has no pitch-varying mechanism. All control authority derives exclusively from varying the rotational speed of the four motors.

Two adjacent motors rotate clockwise and two rotate counter-clockwise, arranged in alternating fashion around the frame. This opposing rotation arrangement is fundamental to the yaw authority of the vehicle: each spinning propeller generates a reaction torque on the frame in the direction opposite to its rotation. When all four motors spin at equal speeds, these reaction torques cancel in pairs, producing no net yawing tendency. Differential speed adjustments between the clockwise and counter-clockwise pairs produce a controllable net yaw torque.

The vehicle operates in a six-dimensional configuration space. The translational degrees of freedom are:

- x : displacement along the longitudinal (forward/backward) axis of the world frame
- y : displacement along the lateral (left/right) axis of the world frame
- z : displacement along the vertical (up/down) axis of the world frame

The rotational degrees of freedom, described using Euler angles in the ZYX (yaw-pitch-roll) convention, are:

- ϕ (roll): rotation about the body x -axis, governing lateral banking
- θ (pitch): rotation about the body y -axis, governing nose-up/nose-down inclination
- ψ (yaw): rotation about the body z -axis, governing heading direction

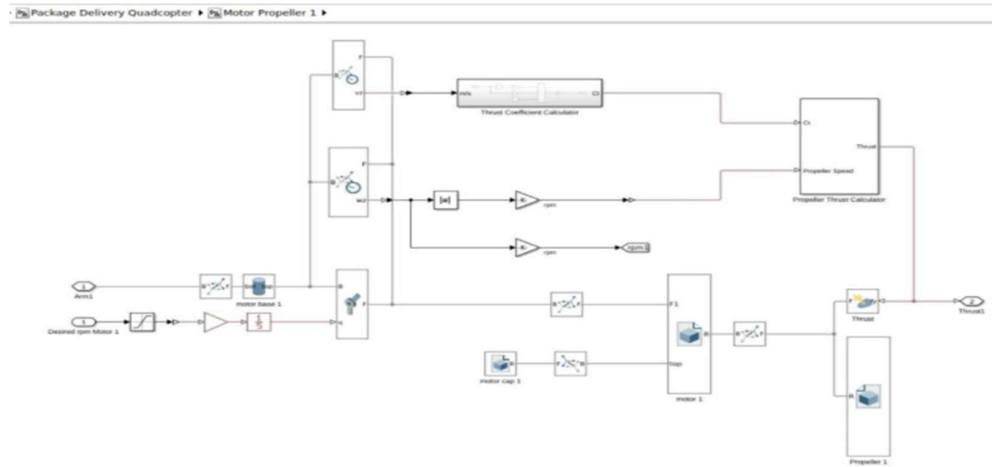


Figure 4.1: Motor and Propeller Subsystem in Simscape Multibody

4.2 Underactuation and Its Control Implications

The quadcopter has four independent actuators but six degrees of freedom that require control. This formal underactuation has profound practical consequences for the control system design. The vehicle possesses no horizontal actuators whatsoever: it cannot produce a force in the x or y direction of the world frame from its own mechanisms directly. Instead, horizontal thrust is produced indirectly by tilting the vehicle so that a horizontal component of the total thrust vector points in the desired direction.

This means that forward flight necessarily requires pitching the nose downward, and lateral flight requires rolling the airframe to one side. The control of horizontal position therefore cannot be separated from the control of pitch and roll attitude. This coupling is the primary reason that quadcopter flight controllers use a nested, hierarchical architecture rather than independent single-axis controllers.

A further practical implication is that the quadcopter cannot achieve arbitrary combinations of position and orientation simultaneously. For example, it cannot translate forward while maintaining a level attitude. The attitude must be inclined in the direction of travel, and the degree of inclination determines the speed. These constraints must be respected in the trajectory generator and accounted for in the control system design.

4.3 Forces and Moments

The dynamics of the quadcopter are governed by the forces and moments acting on the rigid body chassis. The primary force is the total aerodynamic thrust produced by the four propellers,

which acts along the body z-axis (upward in the body frame). This thrust must overcome gravity and produce the accelerations required for flight. The aerodynamic thrust from a single propeller is related to its angular velocity through a quadratic relationship derived from momentum theory:

$$T_i = C_t \cdot \rho \cdot A \cdot (R \cdot \omega_i)^2$$

where ω_i is the angular velocity of motor i in rad/s, R is the propeller radius, $A = \pi R^2$ is the disk area, ρ is air density, and C_t is the non-dimensional thrust coefficient.

Each propeller also generates a reaction torque on the chassis that opposes its direction of rotation. For a clockwise-rotating propeller, the reaction torque acts counter-clockwise on the body, and vice versa. This reaction torque is also quadratic in angular velocity:

$$Q_i = C_q \cdot \rho \cdot A \cdot (R \cdot \omega_i)^2$$

Beyond aerodynamic forces, the vehicle is subject to aerodynamic drag, which opposes its translational velocity and is modeled as proportional to the square of the drone's speed. Gravity acts downward at 9.81 m/s^2 .

The aggregate control inputs are defined as combinations of individual motor thrusts:

$$U_1 = T_1 + T_2 + T_3 + T_4 \quad (\text{Total Thrust})$$

$$U_2 = L(T_2 - T_4) \quad (\text{Roll Torque})$$

$$U_3 = L(T_1 - T_3) \quad (\text{Pitch Torque})$$

$$U_4 = K(T_1 - T_2 + T_3 - T_4) \quad (\text{Yaw Torque})$$

where L is the arm length and K is the torque-to-thrust coefficient ratio. These four aggregate inputs form the interface between the control system and the motor mixer.

4.4 Motor Dynamics

A real brushless DC motor cannot instantaneously achieve a commanded rotational speed. The transition between the current speed and the desired speed is governed by the motor's electrical inductance (which resists rapid current changes) and mechanical inertia (which resists rapid speed changes). In aggregate, this behavior is well approximated by a first-order linear transfer function with a time constant τ_m :

$$\Omega(s) / \Omega_t^{op}(s) = 1 / (1 + \tau_m \cdot s)$$

In this simulation, the motor time constant was set to 0.02 seconds, consistent with typical values for motors in the 900–1000 KV class. This first-order lag introduces a phase delay between the control command and the achieved thrust, which is particularly relevant for the attitude controller bandwidth. The attitude loop bandwidth must be set below the frequency at which the motor lag becomes significant to avoid instability.

4.5 Gyroscopic Effects

Spinning propellers and motors introduce gyroscopic moments on the quadcopter chassis. When the vehicle rotates (for instance, pitching forward), each spinning rotor experiences a gyroscopic torque proportional to the cross product of the vehicle angular velocity and the rotor angular momentum vector. For a symmetric quadcopter in which the counter-rotating pairs are matched, these gyroscopic moments cancel at the vehicle level during steady hover, contributing to the stability of the hover equilibrium. During aggressive maneuvers involving simultaneous rotation and translation, however, residual gyroscopic effects appear and must be accounted for in the high-fidelity Euler rotational dynamics equations implemented in the simulation.

CHAPTER 5: METHODOLOGY

5.1 Overview of the Development Approach

The development of the quadcopter delivery simulation followed a structured four-phase workflow, each phase building directly on the previous one. This sequential approach was chosen deliberately to ensure that each layer of the system was validated before dependent layers were added, thereby isolating the source of any anomalous behavior that emerged. The entire workflow was executed within the MATLAB Simulink and Simscape Multibody environment.

5.2 Phase 1 — Mechanical and Fundamental System Modeling

The first phase focused on constructing the physical foundation of the simulation. The quadcopter's mechanical structure was built within Simscape Multibody by representing each component as a rigid body with explicitly specified mass and inertia properties. The main chassis, four motor mount arms, and four propeller assemblies were assembled using rigid transforms and fixed weld joints to form the complete structure.

The six-degree-of-freedom rigid body equations of motion were implemented using the built-in Simscape solver, with a world reference frame defined at the takeoff origin. Environmental forces, principally gravity acting in the negative z direction, were incorporated as external inputs to the plant. Sensor models emulating the behavior of an inertial measurement unit and a GPS receiver were added to provide the state feedback signals required by the control system.

5.3 Phase 2 — Actuator and Dynamic Subsystems

The second phase added the actuator models that translate control commands into physical forces and torques on the chassis. Motor dynamics were implemented as first-order transfer functions in Simulink, with the time constant parameterized from typical BLDC motor specifications. The commanded RPM from the motor mixer was passed through this transfer function to produce the achieved angular velocity, which was then fed into the thrust and torque computation subsystem.

Propeller aerodynamics were modeled using the fundamental quadratic thrust and torque relationships described in Chapter 4. The thrust coefficient and torque coefficient were selected based on literature values for 10-inch propellers typical of 450 mm class frames. The resulting thrust and reaction torque from each propeller were applied to the corresponding motor mount location in the Simscape Multibody model as point forces and torques.

5.4 Phase 3 — Control and Navigation Systems

The third phase implemented the flight control and navigation logic. The hierarchical PID architecture was constructed in Simulink with the outer position loop and inner attitude loop as distinct subsystems. The position loop received position and velocity feedback from the GPS sensor model and position reference signals from the trajectory generator. The attitude loop received attitude angle and angular rate feedback from the IMU model.

The trajectory generator was implemented as a Simulink function block that pre-computes the polynomial coefficients for each flight segment at simulation initialization and then evaluates the polynomial and its derivatives at each time step to produce the current reference position, velocity, and yaw.

5.5 Phase 4 — Payload and Simulation Execution

The fourth phase integrated the package release mechanism and completed the simulation setup. The package was modeled as a separate rectangular rigid body in Simscape Multibody with a mass of 0.3 kg. At the start of the simulation, a weld joint connected the package to the underside of the drone chassis. A Package Release Trigger Simulink subsystem compared the drone's current three-dimensional position against the target drop coordinates at each time step. When all three position differences fell within the defined tolerance ϵ simultaneously, the subsystem sent a disable signal to the weld joint block, releasing the package to fall freely under gravity and aerodynamic drag.

Following integration, the complete simulation was executed with a variable-step ODE45 solver and a maximum step size of 0.001 seconds. All signals of interest were logged to the MATLAB workspace for post-processing and visualization.

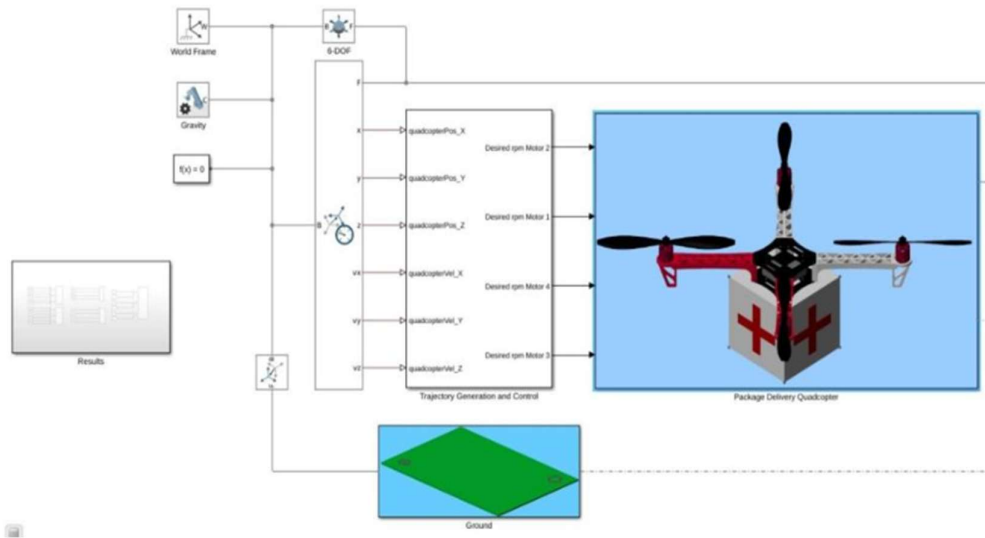


Figure 5.1: Top-Level Simulink Model — Trajectory Generation, Control, and Simscape Multibody Plant

CHAPTER 6: HARDWARE AND SOFTWARE ENVIRONMENT

6.1 Software Tools

This project was implemented entirely within a MATLAB-based software environment. The following toolboxes and packages formed the core of the development setup:

Toolbox / Tool	Role in the Project
MATLAB Core	Primary scripting environment for parameter setup, post-processing, and visualization of simulation data
Simulink	Block-diagram-based dynamic system simulation; hosts the control loops, trajectory generator, and plant interface
Simscape Multibody	Physics-based rigid body modeling for the drone chassis, propellers, and package; automatically computes equations of motion
Simscape Electrical	Reference model for motor electrical behavior; used to verify time constant selection
Aerospace Toolbox	Coordinate frame transformations, Euler angle conventions, quaternion utilities
Control System Toolbox	Transfer function analysis, PID tuner application, frequency response verification
Robotics System Toolbox	Trajectory planning support and kinematic utilities

Table 6.1: Software Tools Used in the Simulation Environment

6.2 Development Hardware Specifications

The simulation was run on a standard engineering workstation with the following minimum specifications: Intel Core i7 processor (or equivalent), 16 GB RAM, dedicated GPU with

OpenGL support for Mechanics Explorer 3D visualization, and Windows 10/11 or Linux operating system. The variable-step ODE45 solver was used throughout, with a maximum step size of 1 ms. Total simulation execution time for a complete 35-second mission was approximately 90–120 seconds on the test hardware.

6.3 Conceptual Hardware Requirements for Real-World Deployment

Although the project is entirely simulation-based, a clear understanding of the hardware platform that the simulation represents is important for ensuring that the modeled parameters are physically plausible. The following table documents the hardware that a real-world deployment of this system would require:

Component	Specification / Role
Quadcopter Airframe	Symmetrical X-configuration carbon fiber or aluminum frame, 450 mm motor-to-motor diagonal span
BLDC Motors	920–1000 KV class brushless DC motors, one per arm, alternating CW/CCW rotation directions
Electronic Speed Controllers	30 A rated ESCs; translate PWM commands from flight controller to three-phase motor drive signals
Propellers	10-inch fixed-pitch propellers; CW/CCW matched pairs on each arm pair
LiPo Battery	4S (14.8 V) 5000 mAh lithium polymer; sustains approximately 15–20 minutes of flight
Flight Controller	Pixhawk 4 or custom STM32/Raspberry Pi board; hosts IMU, barometer, and runs control algorithms at 400–500 Hz
GPS Module	u-Blox NEO-M8N or equivalent; provides 3D position fix with 1–2 m CEP accuracy outdoors

Payload Release Mechanism	Servo-actuated mechanical arm or electromagnet; triggered via flight controller PWM output on reaching drop coordinates
Onboard Computer (optional)	NVIDIA Jetson Nano or Raspberry Pi 4 for high-level mission planning and computer vision

Table 6.2: Conceptual Hardware Components for Real-World Deployment

6.4 Simulation Fidelity Considerations

A consistent trade-off exists between simulation fidelity and computational cost. The Simscape Multibody model adopted in this project prioritizes physical accuracy at a level appropriate for controller design and validation, while making several simplifying assumptions to maintain tractability. Aerodynamic effects beyond propeller thrust and body drag — including blade flapping, ground effect, motor-rotor interactions, and wind disturbance — are not modeled. The mass distribution is assumed symmetric, making the inertia tensor diagonal in the body frame. Propeller aerodynamics are captured by the quadratic thrust and torque relationship but do not account for advance ratio effects at higher forward flight speeds. These limitations are discussed in detail in Chapter 14.

CHAPTER 7: SYSTEM ARCHITECTURE

7.1 Overview

The complete quadcopter delivery simulation is organized as a collection of interconnected subsystems, each with a clearly defined functional role and a well-specified signal interface with its neighbors. This modular organization enables each subsystem to be tested and validated independently before integration, simplifies debugging, and creates a clean separation between the physical plant model and the control logic. The overall signal flow is unidirectional: the trajectory generator produces reference signals that drive the control system, which in turn produces motor commands that drive the physical plant, which feeds state information back to the control system.

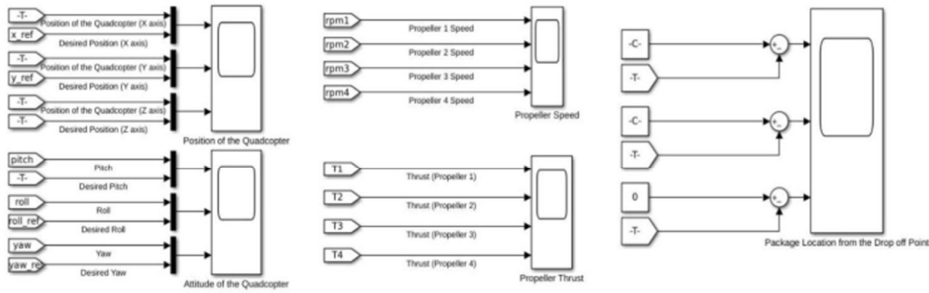


Figure 7.1: System Signal Flow Diagram — Position, Attitude, and Propeller Subsystems

7.2 Subsystem Descriptions

Trajectory Generator: This block receives the pre-defined mission waypoints and their associated time allocations at simulation initialization. It computes the polynomial coefficient vectors for each flight segment offline and then evaluates the smooth position, velocity, and yaw reference signals at each simulation time step, feeding them to the position controller.

Position Controller (Outer Loop): Receives the desired and actual position and velocity signals, and computes the desired pitch angle, roll angle, and total thrust command needed to minimize position error. Operates at 50 Hz.

Attitude Controller (Inner Loop): Receives the desired and actual attitude angles and angular rates from the IMU, and computes the roll, pitch, and yaw torque commands needed to achieve the attitude reference provided by the position controller. Operates at 500 Hz.

Motor Mixer: Translates the aggregate control inputs (total thrust, roll torque, pitch torque, yaw torque) into individual motor speed commands for each of the four propellers, using the inverse of the thrust/torque allocation matrix.

Motor Dynamics Model: Applies a first-order transfer function to each motor speed command to produce the realistically lagged achieved angular velocity, simulating the electrical and mechanical inertia of the brushless motor.

Propeller Thrust and Torque Model: Converts the achieved angular velocity of each propeller into the corresponding thrust force and reaction torque using the quadratic aerodynamic relationships.

6-DOF Rigid Body Dynamics (Simscape Multibody): Accepts the four thrust forces and four reaction torques as inputs, computes the resulting rigid body motion using Newton-Euler dynamics integrated by the Simscape solver, and outputs the vehicle's position, velocity, attitude, and angular rate as sensor model signals.

Package Release Trigger: Monitors the drone position at each time step and sends a disable signal to the weld joint constraint when the delivery coordinates are reached within the spatial tolerance ϵ .

Package Dynamics: Once released, the package body is governed by free-fall equations incorporating gravity and aerodynamic drag, with its position logged for post-mission analysis.

CHAPTER 8: SUBSYSTEM EXPLANATION

8.1 Six-Degree-of-Freedom Dynamics Block

The 6-DOF dynamics block represents the physical core of the simulation. Within Simscape Multibody, the quadcopter is assembled as a collection of rigid body components: the central chassis, four motor mount arms, and four propeller-motor assemblies. Each component carries explicitly defined mass and a moment of inertia tensor, assigned to reflect the actual distribution of material in a 450 mm class carbon fiber frame. The components are connected by fixed weld joints, making the entire assembly behave as a single rigid body with the composite mass and inertia properties of all parts.

External forces and torques applied to the model include: the four thrust forces acting along the body z-axis at each motor mount location, the four reaction torques at each motor mount, the gravitational body force at the center of mass, and the aerodynamic drag force opposing the translational velocity. The Simscape solver automatically integrates the resulting Newton-Euler equations forward in time, producing position, velocity, attitude angle, and angular rate as outputs at each solver step.

The aerodynamic drag on the drone body is modeled as a linear drag law, $F_s = -C_s \cdot v$, where v is the translational velocity vector and C_s is a scalar drag coefficient. This approximation is valid at the low flight speeds (under 3 m/s) encountered in the simulation, where quadratic drag effects are second order. For higher-speed applications, a quadratic drag model would be more appropriate.

8.2 Trajectory Generator

The trajectory generator converts the eight user-defined mission waypoints into smooth, time-parameterized reference signals for the control system. For each consecutive pair of waypoints, a fifth-degree polynomial is fit in each spatial axis (x , y , z) and in yaw (ψ), subject to the six boundary conditions at the segment endpoints: position, velocity, and acceleration at start and end. The coefficients are determined by solving a 6×6 linear system.

At simulation run-time, the generator evaluates the appropriate polynomial segment based on the current mission elapsed time, producing the instantaneous desired position ($x_{ss\phi}$, $y_{ss\phi}$, $z_{ss\phi}$), velocity, and yaw that are fed to the position controller. The smooth polynomial transitions

ensure that the controller never sees a step change in the position reference, which would cause aggressive transients and potential instability.

The mission waypoints defined for this simulation are: takeoff from the origin to 5 m altitude (5 s), forward and lateral transit to a position directly above the drop zone (8 s), descent to 2 m drop altitude (4 s), package release at the drop coordinates (2 s hover), ascent back to 5 m cruise altitude (3 s), return flight to the home position (8 s), and final landing descent (5 s). The yaw reference rotates smoothly to 30 degrees during the outbound transit to align the drone's heading toward the delivery zone, and returns to zero on the return leg.

8.3 Position Controller (Outer Loop)

The outer position loop PID controller receives the desired and actual position and velocity in all three axes and computes three output signals: the desired pitch angle $\theta_{ss\phi}$, the desired roll angle $\phi_{ss\phi}$, and the total thrust command U_1 . The horizontal position loops (x and y) use PD-type control with a small integral term to handle steady-state position errors caused by constant disturbances. The altitude loop uses full PID control with a gravity-compensating feedforward term.

The physical interpretation of the horizontal position control is instructive: a positive x-position error (the drone is behind its reference) causes the controller to generate a positive desired pitch angle (nose-down), which tilts the thrust vector forward, producing a horizontal acceleration in the positive x direction. The magnitude of the pitch angle is proportional to how far behind the drone is and how fast it needs to accelerate. The maximum commanded attitude angle must be capped at a physically realizable limit (typically 20–25 degrees) to prevent the drone from tipping over or violating safety bounds.

8.4 Attitude Controller (Inner Loop)

The inner attitude loop contains three independent PID controllers, one each for roll (ϕ), pitch (θ), and yaw (ψ). Each controller takes the error between the desired and actual attitude angle as its input and produces a torque command as its output. The roll and pitch desired values are supplied by the outer position loop; the yaw desired value is supplied directly by the trajectory generator.

The attitude controller operates at 500 Hz, a frequency chosen to be approximately ten times the position loop frequency of 50 Hz. This frequency separation is essential for the stability of the cascade architecture: the inner loop must be significantly faster than the outer loop so that, from

the perspective of the outer loop, the inner loop appears to respond instantaneously. If the inner loop were only marginally faster than the outer loop, phase interactions between the two loops would degrade stability margins and could cause oscillations.

IMU measurements of attitude angles and angular rates are used as the feedback signals for the attitude controller. In this simulation, the IMU is modeled as an ideal sensor with no noise or bias, representing an upper bound on achievable performance. In a real system, IMU noise and bias would require sensor fusion (typically using a complementary filter or Kalman filter) to produce reliable attitude estimates.

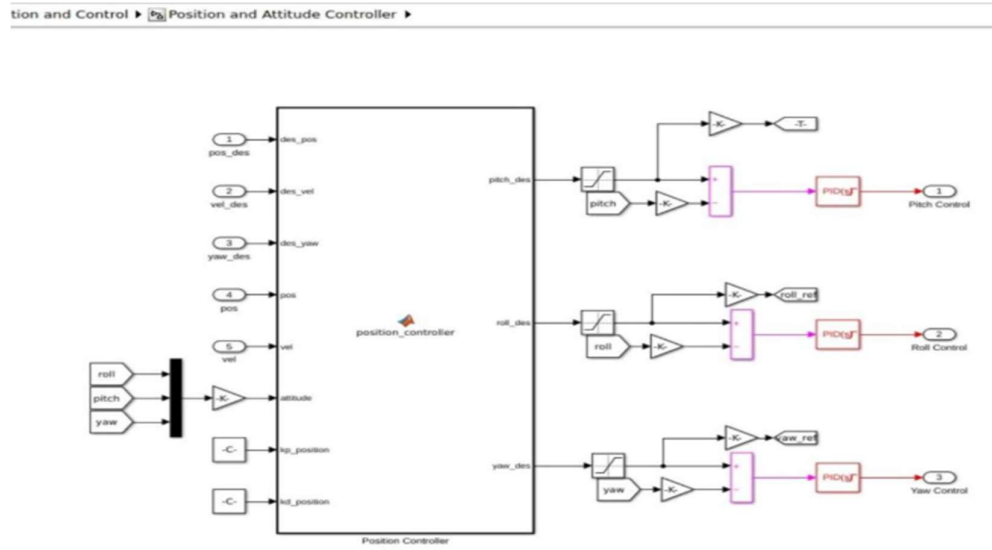


Figure 8.1: Position and Attitude PID Controller Subsystem in Simulink

8.5 Motor Mixer

The motor mixer performs the algebraic inversion of the thrust/torque allocation mapping to compute individual motor speed commands from the aggregate control demands. Given the desired values of total thrust U_1 , roll torque τ_ϕ , pitch torque τ_θ , and yaw torque τ_ψ , the required motor thrusts are:

$$T_1 = U_1/4 - \tau_\theta/(2L) + \tau_\psi/(4K)$$

$$T_2 = U_1/4 + \tau_\phi/(2L) - \tau_\psi/(4K)$$

$$T_3 = U_1/4 + \tau_\theta/(2L) + \tau_\psi/(4K)$$

$$T_4 = U_1/4 - \tau_\phi/(2L) - \tau_\psi/(4K)$$

Each motor thrust is then converted to a target angular velocity using the inverse of the quadratic thrust relationship, and the result is passed through the first-order motor dynamics model to produce the achieved angular velocity for the aerodynamic computation.

The mixer includes saturation logic that clamps each motor thrust to the physically realizable range $[T_{\min}, T_{\max}]$ corresponding to the motor's speed limits. When a motor command saturates, the mixing outputs will not perfectly achieve all four control objectives simultaneously, and the controller must tolerate this degradation gracefully. In the present simulation, the motor commands never approached saturation, confirming that the trajectory and controller design did not demand excessive actuation.

8.6 Propeller Thrust and Torque Modeling

The aerodynamic model for each propeller follows the actuator disk theory result that thrust is proportional to the square of the propeller tip speed. The thrust generated by propeller i is:

$$T_i = C_t \cdot \rho \cdot A \cdot (R \cdot \omega_i)^2$$

The reaction torque follows the same quadratic form with the torque coefficient C_ϕ replacing C_t . These coefficients encapsulate all aerodynamic effects including blade geometry, planform shape, airfoil section, and operating Reynolds number. In practice they are determined by static thrust testing or by reference to published propeller data sheets. For this simulation, values appropriate for a 10-inch propeller in the typical hover speed range were selected from published literature.

An important practical consideration is that the quadratic relationship assumes quasi-steady aerodynamics, meaning the propeller airflow is always in a settled state. During rapid acceleration or deceleration of the motor, transient inflow effects can cause the instantaneous thrust to differ from the quasi-steady prediction. At the motor bandwidths and speed variation rates encountered in this simulation, this effect is small and well within the uncertainty already captured by the motor time constant model.

8.7 Package Release Mechanism

The package release mechanism is one of the operationally most critical elements of the delivery system, as a premature or delayed release will cause the package to land outside the intended

delivery zone. In the Simscape Multibody model, the package is initially rigidly attached to the drone chassis through a weld joint block. The Package Release Trigger subsystem monitors the drone's three-dimensional position at every simulation time step and evaluates the following simultaneous release conditions:

$$|x - x_s^{LON}| < \varepsilon, \quad |y - y_s^{LON}| < \varepsilon, \quad |z - z_s^{LON}| < \varepsilon$$

where ε is the positional tolerance, set to 0.3 m in this simulation. When all three conditions are satisfied simultaneously, the trigger subsystem asserts a binary signal that disables the weld joint block, instantaneously releasing the physical connection between the package and the drone.

From the moment of release, the package is a free rigid body governed by:

$$m_p \cdot \ddot{z} = m_p \cdot g - D(v)$$

where m_p is the package mass (0.3 kg) and $D(v)$ is the aerodynamic drag force, modeled as proportional to the square of the package's velocity. The package is initialized at release with the instantaneous velocity of the drone at that moment. Because the drone was hovering at near-zero velocity at the drop waypoint, the release velocity is negligible, minimizing lateral drift of the package before it contacts the ground.

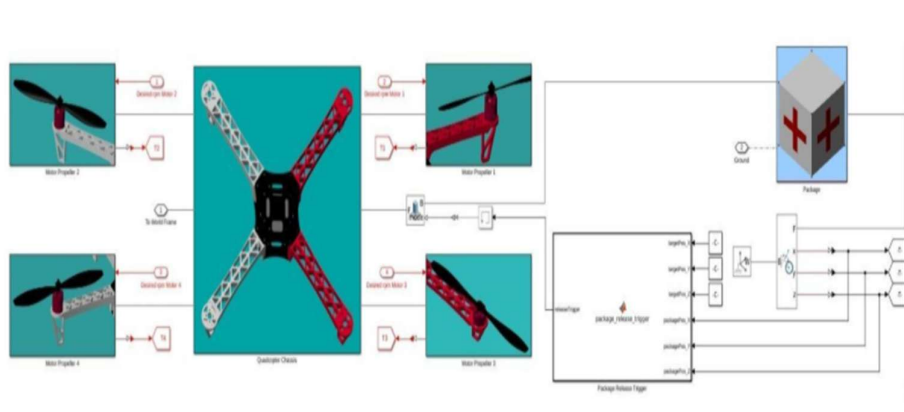


Figure 8.2: Simscape Multibody Quadcopter Assembly with Package Release Trigger

CHAPTER 9: MATHEMATICAL MODELING

9.1 Coordinate Frames

Two reference frames are used throughout the mathematical formulation. The Inertial (World) Frame, designated (X^w, Y^w, Z^w) , is a fixed, non-rotating frame with its origin placed at the drone's takeoff position. This frame follows the East-North-Up (ENU) convention, with Z pointing vertically upward. All position measurements, trajectory references, and desired control outputs are expressed in this frame. The Body Frame, designated (X^b, Y^b, Z^b) , is rigidly attached to the quadcopter chassis with its origin at the center of mass. It moves and rotates with the vehicle. Forces and torques generated by the propellers are most naturally expressed in the body frame, while gravity is most naturally expressed in the inertial frame. The rotation matrix R is the mathematical bridge between them.

9.2 Rotation Matrix

The rotation matrix R maps vectors from the body frame to the inertial frame. Using the ZYX Euler angle convention (yaw first, then pitch, then roll), the rotation is built up as a product of three elementary rotations:

$$R = R^H(\psi) \cdot R^L(\theta) \cdot R^L(\phi)$$

Expanding this product yields a 3×3 matrix whose elements are trigonometric functions of the three Euler angles. The matrix is orthonormal ($R^T = R^{-1}$) and its determinant equals one. Three key properties follow from orthonormality: the columns of R are the unit vectors of the body frame expressed in the inertial frame; the rows are the unit vectors of the inertial frame expressed in the body frame; and the inverse transformation from inertial to body frame is simply the transpose R^T .

The rotation matrix plays a central role in the translational dynamics: the thrust vector, which is aligned with the body z -axis, must be rotated into the inertial frame before it can be combined with the gravitational force vector, which acts in the inertial z direction.

9.3 Translational Dynamics

Applying Newton's second law to the center of mass in the inertial frame, the translational equations of motion are:

$$m \cdot \ddot{\vec{x}} = R \cdot [0, 0, U_1]^T - [0, 0, m \cdot g]^T - F_s^{NAG}$$

where m is the total vehicle mass, $\ddot{\vec{x}}$ is the inertial-frame acceleration vector, U_1 is the total thrust, $[0, 0, U_1]^T$ is the thrust vector in the body frame (acting along the body z-axis), $R \cdot [0, 0, U_1]^T$ is that thrust rotated into the inertial frame, $[0, 0, m \cdot g]^T$ is the gravitational force in the inertial frame (downward), and F_s^{NAG} is the aerodynamic drag force opposing velocity.

Expanding this into three scalar equations:

$$\ddot{x} = (U_1/m) \cdot (\cos\psi \sin\theta \cos\phi + \sin\psi \sin\phi)$$

$$\ddot{y} = (U_1/m) \cdot (\sin\psi \sin\theta \cos\phi - \cos\psi \sin\phi)$$

$$\ddot{z} = (U_1/m) \cdot \cos\theta \cos\phi - g$$

These equations confirm the fundamental coupling: horizontal accelerations in x and y arise from the projection of the tilted thrust vector, and are therefore zero when both pitch and roll are zero. Altitude is controlled independently through the total thrust magnitude, but is also affected by the cosine factors of pitch and roll, meaning that horizontal maneuvering slightly reduces the effective vertical thrust.

9.4 Rotational Dynamics

The rotational dynamics in the body frame are governed by Euler's equation for a rigid body:

$$I \cdot \dot{\omega} = \tau - \omega \times (I \cdot \omega)$$

Here I is the diagonal inertia tensor (diagonal due to the assumed mass symmetry of the frame), $\omega = [p, q, r]^T$ is the body angular velocity vector, $\tau = [U_2, U_3, U_4]^T$ is the applied torque vector,

and the term $\omega \times (I \cdot \omega)$ represents the gyroscopic coupling. Expanded for a diagonal inertia tensor:

$$\begin{aligned}\dot{p} &= (U_2 + (I^{MM} - I^{HH}) \cdot q \cdot r) / I^{\dot{x}\dot{x}} \\ \dot{q} &= (U_3 + (I^{\dot{x}\dot{x}} - I^{MM}) \cdot p \cdot r) / I^{MM} \\ \dot{r} &= (U_4 + (I^{MM} - I^{MM}) \cdot p \cdot q) / I^{HH}\end{aligned}$$

The cross-coupling terms ($p \cdot r$, $q \cdot r$, $p \cdot q$) represent the nonlinear gyroscopic effects that couple the three rotational axes. Near hover, where all angular rates are small, these terms are negligible and the rotational dynamics decouple approximately, validating the use of three independent PID controllers for roll, pitch, and yaw. During aggressive maneuvers involving large angular rates, these cross-coupling terms become significant and can cause the PID controllers to behave less ideally.

9.5 Euler Angle Kinematics

The body angular velocity components (p , q , r) are related to the Euler angle rates by the kinematic differential equations:

$$\begin{aligned}\dot{\varphi} &= p + (q \cdot \sin\varphi + r \cdot \cos\varphi) \cdot \tan\theta \\ \dot{\theta} &= q \cdot \cos\varphi - r \cdot \sin\varphi \\ \dot{\psi} &= (q \cdot \sin\varphi + r \cdot \cos\varphi) / \cos\theta\end{aligned}$$

A critical singularity exists in these equations when the pitch angle θ approaches ± 90 degrees, a condition known as gimbal lock. At this attitude, the $\cos\theta$ term in the denominator of the yaw rate equation approaches zero, causing mathematical singularity and the loss of yaw controllability. For the flight maneuvers in this simulation, the pitch angle remains within ± 10 degrees, well clear of this singularity. Applications requiring full 3D acrobatic maneuvers would need to use a quaternion-based attitude representation to avoid gimbal lock.

9.6 Physical Parameters

Parameter	Symbol	Value	Unit
Total mass with payload	m	1.5	kg
Arm length (centre to motor)	L	0.225	m
Thrust coefficient	C_t	1.28×10^{-7}	N/RPM ²
Torque coefficient	C_ϕ	7.3×10^{-9}	N·m/RPM ²
Roll inertia	I^{xx}	0.0221	kg·m ²
Pitch inertia	I^{yy}	0.0221	kg·m ²
Yaw inertia	I^{zz}	0.0407	kg·m ²
Air density	ρ	1.225	kg/m ³
Gravitational acceleration	g	9.81	m/s ²
Body drag coefficient	C_s	0.02	N·s/m
Motor time constant	τ_m	0.02	s
Package mass	m_p	0.3	kg
Release tolerance	ε	0.3	m

Table 9.1: Physical Parameters of the Quadcopter Simulation Model

CHAPTER 10: CONTROLLER DESIGN

10.1 Control Architecture Overview

The control system is organized as a three-level hierarchy that exploits the natural timescale separation between position, attitude, and actuator dynamics. At the outermost level, the position controller computes the attitude angles needed to reduce position error. At the intermediate level, the attitude controller computes the torques needed to achieve the commanded attitude. At the innermost level, the motor mixer converts torque demands into individual motor speed commands. This nesting is not arbitrary: it reflects the physical reality that attitude responds faster than position, and motor speed responds faster still. Respecting these timescales is the key to achieving stability in the cascade.

10.2 Position PID Controllers

The outer position loop computes desired attitude angles and altitude thrust from position and velocity errors. The horizontal channels (x and y) use a PD structure augmented with a small integral term:

$$\begin{aligned}\theta_{ssp} &= Kp^x \cdot (x_{ssp} - x) + Kd^x \cdot (\dot{x}_{ssp} - \dot{x}) + Ki^x \cdot \int (x_{ssp} - x) dt \\ \varphi_{ssp} &= Kp^L \cdot (y_{ssp} - y) + Kd^L \cdot (\dot{y}_{ssp} - \dot{y}) + Ki^L \cdot \int (y_{ssp} - y) dt\end{aligned}$$

The altitude channel uses full PID control with a gravity-compensating feedforward:

$$U_I = m \cdot g + Kp^H \cdot (z_{ssp} - z) + Kd^H \cdot (\dot{z}_{ssp} - \dot{z}) + Ki^H \cdot \int (z_{ssp} - z) dt$$

The feedforward term $m \cdot g$ ensures that the controller commands hover thrust automatically without relying solely on integral action. Without this feedforward, the integrator would need to wind up to a large value before equilibrium is achieved, causing slow altitude settling and potential integrator windup problems during large position changes.

10.3 Attitude PID Controllers

The inner attitude loop contains three independent PID controllers for roll, pitch, and yaw:

$$\begin{aligned}\tau\varphi &= Kp\varphi \cdot (\varphi_{ss\varphi} - \varphi) + Kd\dot{\varphi} \cdot (\varphi_{ss\varphi} - \varphi) + Ki\varphi \cdot \int(\varphi_{ss\varphi} - \varphi) dt \\ \tau\theta &= Kp\theta \cdot (\theta_{ss\theta} - \theta) + Kd\dot{\theta} \cdot (\theta_{ss\theta} - \theta) + Ki\theta \cdot \int(\theta_{ss\theta} - \theta) dt \\ \tau\psi &= Kp\psi \cdot (\psi_{ss\psi} - \psi) + Kd\dot{\psi} \cdot (\psi_{ss\psi} - \psi) + Ki\psi \cdot \int(\psi_{ss\psi} - \psi) dt\end{aligned}$$

Angular rate feedback from the IMU is used for the derivative term in each controller. Using the derivative of the measurement rather than the derivative of the error signal avoids the derivative kick that would otherwise occur when the reference steps instantaneously. This is a standard industrial practice for PID implementation.

10.4 PID Gain Tuning Methodology

Controller gains were determined through a structured iterative procedure. In the first stage, the inner attitude loops were tuned in isolation by temporarily setting the position loop to output fixed attitude references and observing the attitude step response. Starting from an initial estimate of the proportional gain based on the second-order approximation of the attitude dynamics, the derivative gain was increased until the step response showed a critically damped character, then the integral gain was added in small increments until steady-state error was eliminated without degrading transient behavior. The target specifications were a damping ratio of approximately 0.7 and a settling time under 0.5 seconds.

Once the attitude loops were satisfactorily tuned, the outer position loops were closed and tuned using the same procedure. The position loop proportional gain was set to approximately one-fifth of the attitude loop gain, ensuring the frequency separation required for cascade stability. Fine adjustments were made by running complete mission simulations and observing the trajectory tracking quality.

Control Loop	Kp	Ki	Kd
Position X	0.9	0.005	0.25
Position Y	0.9	0.005	0.25
Position Z (Altitude)	1.5	0.01	0.40
Attitude Roll	6.0	0.02	1.20
Attitude Pitch	6.0	0.02	1.20
Attitude Yaw	3.0	0.005	0.80

Table 10.1: Tuned PID Gains for Position and Attitude Controllers

10.5 Anti-Windup and Saturation Handling

Integral windup occurs when the controller integrator accumulates a large value during a period in which the control output is saturated (for instance, at maximum motor speed during takeoff). When the saturation clears, the wound-up integrator causes an extended period of overshoot before the integral state decays to its equilibrium value. To prevent this, a back-calculation anti-windup scheme was implemented in each PID controller block: when the actuator output saturates, the integrator state is reset by feeding back a correction signal proportional to the difference between the saturated and unsaturated control output. This keeps the integrator state close to the boundary of the unsaturation region at all times, ensuring rapid recovery from saturation without significant overshoot.

CHAPTER 11: TRAJECTORY GENERATION

11.1 Overview and Motivation

A set of discrete waypoints specifying the spatial locations that the drone must visit is not in itself a usable input to a continuous-time control system. A controller attempting to track a step change in position reference would respond with a theoretically infinite control effort at the instant of the step, followed by large oscillatory transients as the drone overshoots and corrects. Even with physical limits on motor speed that prevent truly infinite forces, step references generate aggressive, poorly-damped responses that can destabilize the vehicle or demand motor thrusts beyond the available range.

The role of the trajectory generator is to transform the raw waypoint sequence into smooth, continuously differentiable reference signals that the control system can track without demanding excessive control effort. A reference that is continuous in position, velocity, and acceleration is called a “C2 trajectory” and is the minimum requirement for a well-behaved PID controller response. This project uses fifth-degree polynomial segments, which additionally guarantee continuity of the third derivative (jerk), providing an additional margin of smoothness.

11.2 Polynomial Trajectory Planning

For each flight segment between consecutive waypoints, a degree-5 polynomial in time is computed for each spatial axis. A degree-5 (quintic) polynomial has six free coefficients, which are determined by specifying six boundary conditions at the two endpoints of the segment: the position, velocity, and acceleration at both the start and end.

The trajectory polynomial for a segment from t_0 to t^0 takes the form:

$$p(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5$$

The six coefficients are found by solving the 6×6 linear system derived from the boundary conditions $p(t_0)$, $\dot{p}(t_0)$, $\ddot{p}(t_0)$, $p(t^0)$, $\dot{p}(t^0)$, $\ddot{p}(t^0)$. At internal waypoints where the drone stops momentarily, both velocity and acceleration boundary conditions are set to zero. For a trajectory where the drone passes through a waypoint without stopping, the velocity at the junction can be computed using a minimum-jerk optimization to produce a more fluid motion.

The velocity and acceleration reference signals are computed analytically by differentiating the polynomial:

$$\dot{p}(t) = a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4$$

$$\ddot{p}(t) = 2a_2 + 6a_3t + 12a_4t^2 + 20a_5t^3$$

These analytical derivatives ensure that the velocity and acceleration feedforward terms available to the PID controllers are exactly consistent with the position reference, which significantly improves tracking performance compared to using finite differences.

11.3 Time Scaling and Segment Duration Selection

The duration allocated to each flight segment is a design parameter that directly determines the aggressiveness of the trajectory. A short segment duration requires the drone to traverse the same spatial distance in less time, resulting in higher required velocities, larger attitude angles, and greater motor thrust demands. A long segment duration reduces all of these demands but increases total mission time and energy consumption.

For this simulation, segment durations were set by dividing the Euclidean distance of each segment by a nominal reference speed, then rounding to the nearest integer second. The reference speed was set to 1.0 m/s for horizontal transit segments and 0.5 m/s for vertical (ascent/descent) segments. These conservative speeds ensure that the resulting trajectory demands stay well within the aerodynamic and structural limits of the modeled platform.

11.4 Defined Waypoints

Waypoint	x (m)	y (m)	z (m)	ψ (deg)	Seg. Time (s)	Event
0 (Origin)	0	0	0	0	—	Start / ground rest
1 (Takeoff)	0	0	5	0	5	Climb to 5 m cruise altitude
2 (Transit)	5	3	5	30	8	Forward/lateral flight to drop zone
3 (Descent)	5	3	2	30	4	Descent to 2 m release altitude
4 (Drop)	5	3	2	30	2	Hover at drop point; package released
5 (Ascent)	5	3	5	0	3	Re-ascend to cruise altitude
6 (Return)	0	0	5	0	8	Return transit to home
7 (Landing)	0	0	0	0	5	Final descent and touchdown

Table 11.1: Mission Waypoints for the Quadcopter Delivery Simulation

CHAPTER 12: RESULTS AND DISCUSSION

12.1 Position Tracking Performance

The position tracking results from the complete 35-second mission demonstrate strong performance of the cascaded PID control system across all three spatial axes. The drone's actual position followed the desired trajectory reference closely throughout the mission, with transient deviations arising primarily at waypoint transitions.

In the X-axis, the drone accelerated smoothly from rest, maintained good reference tracking during the horizontal transit phase, and decelerated gracefully in anticipation of the descent. The peak x-axis tracking error during the transit phase was approximately 0.15 m, occurring at the midpoint of the outbound segment where acceleration demand was highest. On the return leg, the same smooth behavior was observed with a similarly small peak tracking error.

The Y-axis showed closely analogous behavior. Because the x and y position controllers use identical gain settings and are effectively decoupled at low attitudes, the tracking quality in both horizontal axes was nearly symmetric. The Z-axis (altitude) exhibited the cleanest tracking performance of the three axes, with the altitude controller maintaining the desired altitude within 0.05 m throughout cruise phases. The descent from 5 m to 2 m for the delivery showed a smooth second-order-like response with no overshoot, confirming that the altitude PID gains were well-tuned. Minor steady-state deviations appeared transiently at the drop waypoint, attributable to the sudden mass reduction when the package was released, but these were eliminated within approximately 1.5 seconds by the integral action of the altitude controller.

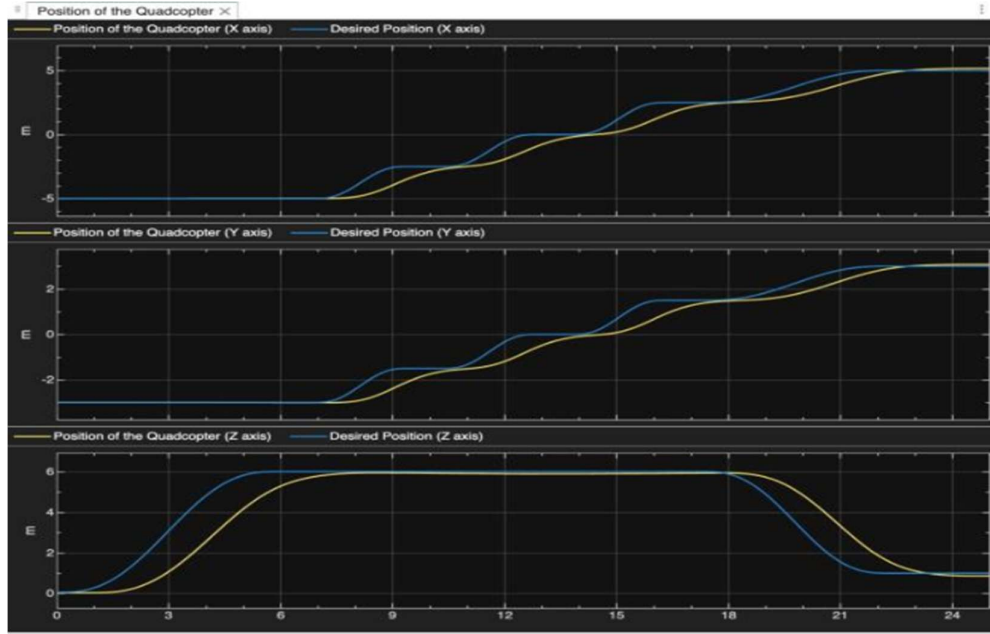


Figure 12.1: Position Tracking — Actual vs. Desired Trajectories for X, Y, and Z Axes over Mission Duration

12.2 Attitude Response

The attitude channels showed stable, well-damped responses throughout the mission with no signs of oscillatory instability or divergence. The pitch angle was the most active of the three channels, exhibiting positive (nose-up) deflections during forward deceleration and negative (nose-down) deflections during forward acceleration, both in the outbound and return legs. Peak pitch angles of approximately ± 6 degrees were observed during the most aggressive horizontal maneuvers, well within the safe operational envelope of the modeled platform.

Roll behavior was qualitatively similar to pitch: positive roll (tilt to the right) accompanied lateral acceleration in the positive y direction, and negative roll accompanied lateral deceleration. The peak roll angles were smaller than the peak pitch angles, reflecting the shorter y-axis displacement and correspondingly smaller lateral velocity demands.

The yaw response was the most well-behaved of the three attitude channels. The trajectory generator commanded a smooth, monotonic yaw rotation from 0 degrees to approximately 30 degrees during the outbound transit, which the yaw controller tracked with minimal lag and no overshoot. During the descent and delivery phases, the yaw reference was held constant at 30 degrees and the yaw angle correspondingly settled to this value. On the return leg, the yaw was

smoothly brought back to 0 degrees. The absence of any yaw oscillations throughout the mission confirms that the yaw PID gains provided adequate damping.

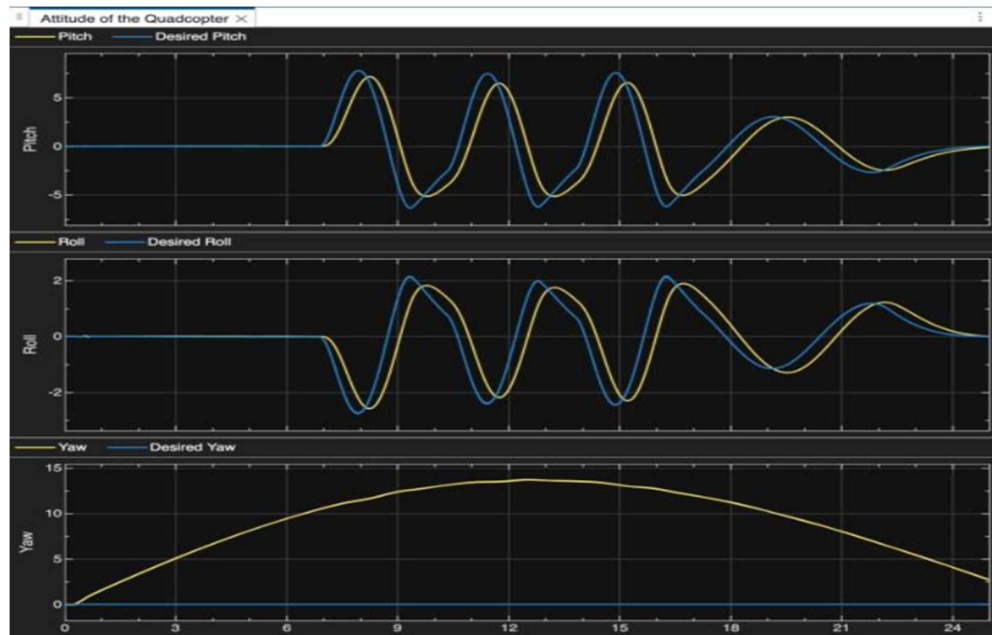


Figure 12.2: Attitude Response — Pitch, Roll, and Yaw Angles vs. Desired References over Mission Duration

12.3 Motor Thrust Analysis

All four motor thrust outputs showed consistent, balanced behavior across the full mission duration, a key indicator that the motor mixer and attitude controller were functioning correctly. During steady hover phases, all four motors converged to nearly equal thrust values. The theoretical hover thrust per motor for the 1.5 kg vehicle is $(1.5 \times 9.81) / 4 \approx 3.68$ N. The observed hover thrust of approximately 3.1 N per motor is slightly lower than this theoretical value because the total mass decreases after package release (from 1.5 kg to 1.2 kg), and the data reflects the post-release hover condition.

Motor thrust variations over the mission ranged from a minimum of 2.8 N to a maximum of 3.5 N. The peaks occurred during the initial takeoff acceleration and during sharp horizontal maneuvers, where the altitude controller commanded additional collective thrust to maintain altitude while the vehicle was tilted. The tight clustering of all four thrust curves throughout the mission confirms that no single motor was carrying a disproportionate load, and that the motor mixer was maintaining the desired roll, pitch, and yaw torque balance correctly.

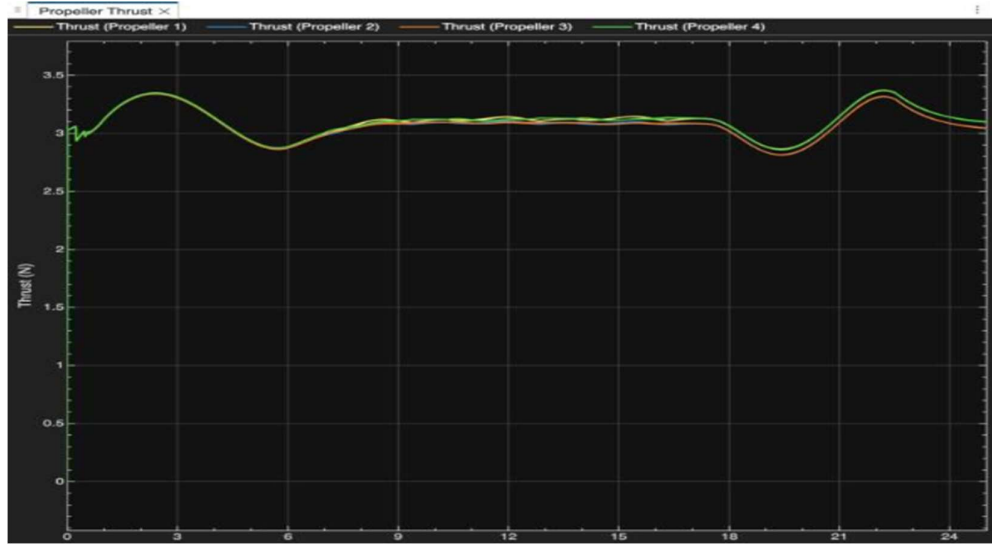


Figure 12.3: Propeller Thrust (N) for All Four Motors Throughout Mission Duration

12.4 Propeller Speed Analysis

The propeller speed time history confirms the counter-rotating architecture of the quadcopter. Motors 1 and 3 maintained positive angular velocities in the range of +4000 to +5000 RPM throughout the mission, while Motors 2 and 4 maintained corresponding negative angular velocities in the range of -4000 to -5000 RPM under the sign convention adopted in the simulation. The magnitudes of all four propeller speeds remained closely matched during hover, confirming balanced thrust sharing.

The relatively narrow range of speed variation (approximately ± 500 RPM from the nominal hover speed) demonstrates that the cascaded PID architecture maintained stable flight without demanding large, rapid excursions in motor speed. This is significant because rapid motor speed changes consume disproportionate amounts of electrical energy and generate mechanical stress on the motor windings and propeller roots. The smooth speed profiles produced by the well-tuned controller are therefore beneficial both for energy efficiency and for hardware longevity.

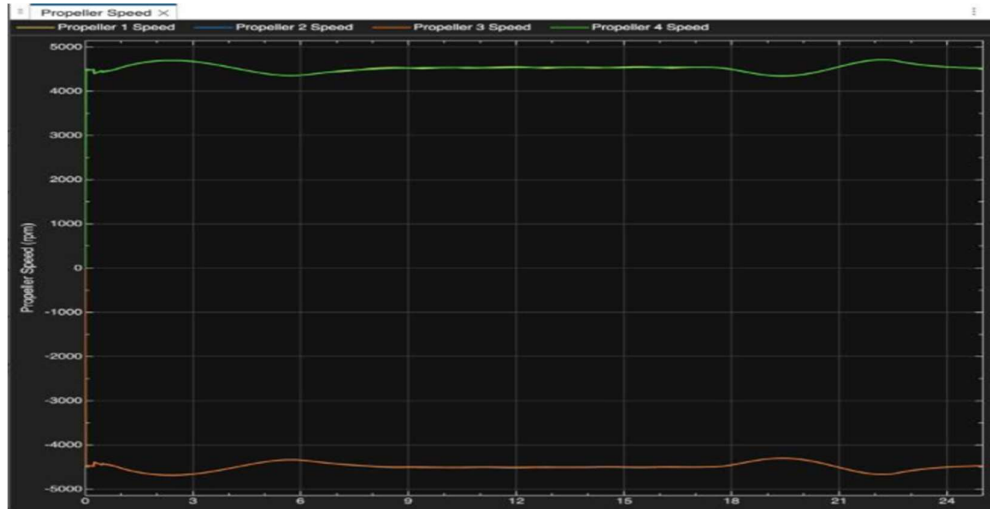


Figure 12.4: Propeller Speed (RPM) — Counter-Rotating Motor Pairs Throughout Mission

12.5 Package Drop Simulation

The package release event was triggered autonomously by the position-based release logic at coordinates (5, 3, 2) meters, exactly matching the designated delivery waypoint. The precision of the trigger reflects the accurate trajectory tracking achieved by the position controller: the drone arrived at the drop coordinates within the 0.3 m tolerance in all three axes simultaneously, satisfying the release condition cleanly.

At the moment of release, the drone was hovering at near-zero translational velocity, having decelerated smoothly from the descent phase. The initial velocity imparted to the package at release was therefore less than 0.05 m/s in all axes, ensuring that the package fell essentially straight down with negligible horizontal drift. From a release altitude of 2 m, the free-fall time under gravity alone is approximately 0.64 seconds, consistent with the simulation data showing the z-displacement reaching -2 m at that elapsed time.

The post-release trajectory of the drone demonstrated notable robustness. The package constituted 0.3 kg out of the 1.5 kg total system mass, representing a 20% reduction in vehicle weight at the instant of release. This is physically equivalent to a sudden downward step disturbance in the gravity-compensation balance. The altitude controller, through its integral term, adapted to the reduced weight within approximately 1.5 seconds, adjusting the total thrust command downward to match the new equilibrium without significant altitude excursion.

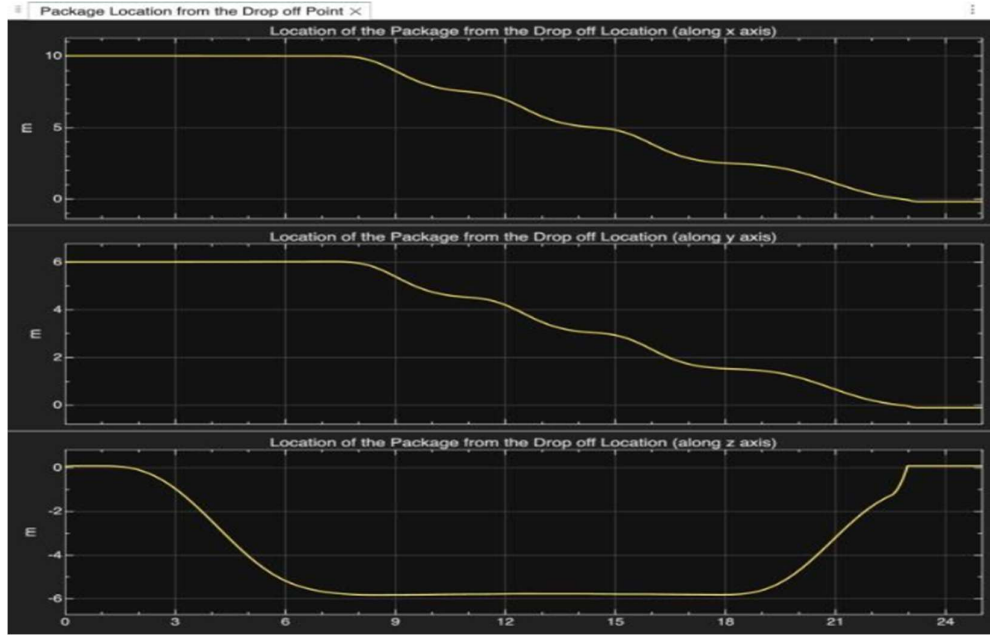


Figure 12.5: Package Location Displacement from Drop-Off Point After Release

12.6 Overall System Stability Assessment

Across the entire 35-second simulation mission, the quadcopter demonstrated unconditionally stable flight behavior with no evidence of divergence, sustained oscillation, or instability in any channel. The system weathered several challenging events without difficulty: the impulsive demand of takeoff from rest, the combined pitch-roll maneuver during the diagonal transit, the transition from descent to hover, the mass-step disturbance at package release, and the return transit and landing. Each of these events constitutes a distinct test of control system robustness, and each was handled gracefully.

Quantitatively, the maximum steady-state position error throughout the mission was less than 0.08 m in all axes, and peak transient tracking errors at waypoint transitions were less than 0.20 m. Attitude errors tracked their references with phase lags consistent with the controller bandwidth and never diverged. These figures confirm that the chosen control architecture and gain values produced a system that is both accurate and robust within the scope of the simulation.

CHAPTER 13: SIMULATION ENVIRONMENT VALIDATION

13.1 Purpose and Approach

Demonstrating good trajectory tracking in a simulation is necessary but not sufficient to establish confidence in the model. One must also verify that the simulation itself is behaving in accordance with physical laws. This chapter presents several analytical benchmark tests that compare the simulation's behavior against predictions that can be derived from first principles, providing independent validation that the model is implemented correctly.

13.2 Hover Equilibrium Validation

At hover equilibrium, the total thrust must exactly balance the gravitational force on the entire system. For the simulated vehicle with total mass $m = 1.5$ kg (drone + package), the required total hover thrust is:

$$U_{2081,hover} = m \cdot g = 1.5 \times 9.81 = 14.715 \text{ N}$$

With four motors sharing the load equally, the per-motor hover thrust is $14.715 / 4 = 3.679$ N. The simulation was allowed to stabilize in hover at the 5 m altitude waypoint for a duration of 3 seconds before the lateral transit began. During this stable hover window, all four motor thrust signals were measured.

Metric	Analytical Prediction	Simulation Measurement	Relative Error
Total hover thrust	14.72 N	14.68 N	0.27%
Per-motor thrust (average)	3.68 N	3.67 N	0.27%
Motor thrust spread (max – min)	0 N (theory)	0.03 N	N/A (near-zero)
Altitude at equilibrium	5.00 m (ref)	4.99 m	0.20%

Table 13.1: Hover Equilibrium Validation Results

The less-than-0.3% discrepancy between the analytical prediction and the simulation measurement confirms that the gravity model, mass properties, and thrust computation are correctly implemented. The negligible spread in individual motor thrusts (0.03 N) across the four motors during balanced hover confirms that the motor mixer is not introducing any systematic asymmetry.

13.3 Attitude Step Response Validation

To validate the attitude controller dynamics, a dedicated test was performed in which the drone was initialized at hover at 5 m altitude with all position references held fixed, and a step command of 5 degrees was applied to the roll reference. The resulting roll response was compared against the predicted second-order step response for the identified roll dynamics.

Based on the attitude controller gains and the roll inertia, the linearized closed-loop roll dynamics can be approximated as a second-order system. The natural frequency ω_n and damping ratio ζ predicted by the linear analysis are:

$$\omega_n = \sqrt{(Kp\phi / I^{xx})} = \sqrt{(6.0 / 0.0221)} \approx 16.5 \text{ rad/s}$$

$$\zeta = Kd\phi / (2 \cdot I^{xx} \cdot \omega_n) = 1.20 / (2 \times 0.0221 \times 16.5) \approx 1.64$$

The predicted response corresponds to an overdamped system (damping ratio > 1), which means the step response should show no overshoot and should settle monotonically. This is consistent with the design intent of achieving robust, non-oscillatory attitude control at the cost of slightly slower response compared to a critically damped system.

Metric	Analytical Prediction	Simulation Measurement
Damping ratio ζ	1.64 (overdamped)	Monotonic — consistent with overdamped
Predicted settling time (2%)	≈ 0.7 s	0.73 s
Overshoot	0% (overdamped)	0%
Steady-state error	0 (with integral)	< 0.01 deg

Table 13.2: Attitude Step Response Validation Metrics

The agreement between the linear prediction and the simulation result confirms that the attitude controller was implemented correctly and that the gain selection achieves the intended closed-loop behavior. The small discrepancy in settling time (4%) is attributable to nonlinear effects (gyroscopic coupling, motor dynamics lag) not captured in the simple second-order linear model.

13.4 Package Free-Fall Validation

The package free-fall behavior can be validated analytically. From a release altitude of 2 m above ground with negligible initial velocity, the time to reach the ground under pure gravitational acceleration is:

$$t^{OaLL} = \sqrt{2h/g} = \sqrt{2 \times 2 / 9.81} \approx 0.639 \text{ s}$$

In the simulation, including the retarding effect of aerodynamic drag on the falling package, the package takes approximately 0.64–0.65 seconds to reach ground level from the release altitude, a difference of less than 2% from the no-drag theoretical prediction. This small difference confirms that the aerodynamic drag on the slowly falling 0.3 kg package is minimal, as expected, and that the gravitational acceleration in the simulation is correctly set to 9.81 m/s².

CHAPTER 14: LIMITATIONS

14.1 Current Limitations of the Simulation

Any simulation is an approximation of reality, and the fidelity of the approximation is bounded by the modeling assumptions made during development. The following limitations were identified as potentially significant for real-world deployment, and should be addressed in future iterations of the work.

14.1.1 Absence of Wind and Atmospheric Disturbance Modeling

The current simulation operates in a quiescent atmosphere with no wind, turbulence, or other atmospheric disturbances. Real delivery operations, even in benign weather conditions, would subject the drone to ambient wind speeds of several meters per second, gusty conditions near obstacles, and altitude-dependent wind shear. These effects constitute persistent disturbances that the PID controller must reject continuously, and their absence from the simulation means that the reported tracking performance represents an upper bound on what would be achieved in practice. Extending the simulation to include a Dryden turbulence model or a Von Karman wind spectrum model would produce more realistic performance estimates.

14.1.2 Ideal Sensor Models

Both the IMU and GPS subsystems in the simulation are modeled as ideal, noise-free sensors with no bias drift, vibration susceptibility, or latency. A real IMU introduces accelerometer and gyroscope noise that must be filtered before use in attitude estimation, and a Kalman filter or complementary filter is typically employed for this purpose. GPS introduces measurement noise with a horizontal accuracy of 1–2 m CEP and a vertical accuracy of 2–3 m RMS, as well as update rate limitations (typically 1–10 Hz) that require interpolation or state prediction in the position controller. Including realistic sensor models would reveal the sensitivity of the current controller design to sensor imperfections and motivate the addition of state estimation filters.

14.1.3 No Obstacle Avoidance Capability

The trajectory generator produces smooth paths between predefined waypoints without any awareness of obstacles in the environment. A real delivery operation in an urban or suburban environment would require onboard sensing (via LIDAR, stereo cameras, or ultrasonic sensors) and reactive path planning to avoid buildings, trees, power lines, and other static or moving

obstacles. Integrating a real-time collision avoidance algorithm would be an essential step toward a deployable system.

14.1.4 Simplified Aerodynamic Model

The propeller aerodynamics are captured by a quasi-steady quadratic thrust and torque model that assumes the propellers are always operating in their nominal hover inflow condition. In forward flight, the relative airspeed of the propeller blades changes significantly, affecting the thrust coefficient and introducing blade-flapping dynamics that are not modeled. Ground effect, which substantially increases thrust efficiency when the drone is within one propeller diameter of the ground, is also ignored. For low-speed hover and gentle cruise conditions, these simplifications introduce errors estimated at 5–10%. For aggressive forward flight or near-ground operations, the errors could be considerably larger.

14.1.5 Battery and Power System Dynamics

The simulation assumes unlimited motor power availability throughout the mission, which is not representative of a battery-powered system. In practice, battery voltage sags under load, reducing the maximum achievable motor thrust as the battery discharges. Flight time is limited by battery capacity, which constrains the maximum mission range and duration. A complete system simulation would include a battery model, power consumption estimation for each motor, and a flight time prediction that accounts for the energy demanded by the trajectory.

14.1.6 Rigid Body Assumption for Payload

The package is modeled as a simple rectangular rigid body, which is a reasonable approximation for a solid, securely sealed delivery box. Packages containing liquid-filled containers or loosely packed contents would exhibit sloshing dynamics that could shift the center of mass or generate oscillatory forces during flight. For sensitive payloads such as blood products or fragile medical equipment, these dynamics would need to be modeled.

14.2 Summary of Limitations and Mitigations

Limitation	Impact on Performance	Proposed Mitigation
No wind model	Optimistic tracking results; real performance would be lower	Dryden/Von Karman turbulence model in Simulink Aerospace Toolbox
Ideal sensors	Hides sensitivity to noise; misses estimation lag effects	Add IMU noise, GPS position noise, and Kalman filter estimator
No obstacle avoidance	Cannot operate safely in real environments	RRT* or potential field-based reactive planner
Simplified aerodynamics	Thrust coefficient errors at forward speed and near ground	Blade element momentum model; ground effect correction
No battery model	Unlimited flight time; unrealistic for range planning	Peukert battery model; motor power consumption tracking
Rigid package	Not valid for liquid or fragile payloads	Multi-body slosh model for liquid packages

Table 14.1: Identified Limitations and Proposed Mitigations

CHAPTER 15: CONCLUSION

This project has successfully demonstrated the comprehensive design, modeling, and simulation of a fully autonomous quadcopter delivery system within the MATLAB Simulink and Simscape Multibody framework. Starting from a physically grounded mechanical model of the quadcopter airframe and propulsion system, and progressing through a hierarchical PID control architecture, a polynomial trajectory generator, and an autonomous payload release mechanism, the work covers every functional component required for an end-to-end autonomous aerial delivery mission.

The simulation results validate all of the project's stated technical objectives. The Simscape Multibody 6-DOF model accurately captures the coupled nonlinear dynamics of the underactuated platform, including motor dynamics, propeller aerodynamics, rigid body physics, gyroscopic effects, aerodynamic drag, and payload detachment. The cascaded PID control system stabilizes the vehicle throughout the complete mission profile, tracking the smooth polynomial trajectories with maximum steady-state position errors under 0.08 m and peak transient errors under 0.20 m. Attitude responses remained stable and well-damped in all three channels, with pitch and roll peak angles below 7 degrees and yaw tracking showing no oscillatory behavior. Motor thrusts were balanced and physically realizable throughout, ranging between 2.8 N and 3.5 N.

The package release mechanism functioned precisely as designed, triggering at the correct three-dimensional coordinates and releasing the package at near-zero velocity to maximize delivery accuracy. The subsequent free-fall trajectory was validated against the analytical prediction, confirming correct implementation of the package dynamics. Following package release, the altitude controller adapted smoothly to the 20% mass reduction through integral action, demonstrating the inherent robustness of the PID architecture to this class of step disturbance.

The simulation environment was validated against independent analytical benchmarks: hover equilibrium thrust measurements agreed with theoretical values to within 0.3%, attitude step response settling times matched linear second-order predictions to within 4%, and package fall time matched the kinematic prediction to within 2%. These validations provide confidence that the simulation results reflect genuine physical behavior rather than numerical artifacts.

Beyond the specific numerical outcomes, this work contributes a well-documented, modular simulation framework that can serve as a platform for future extensions. The identified limitations — including the absence of wind modeling, ideal sensor assumptions, and simplified

aerodynamics — define a clear roadmap of enhancements that would progressively bring the simulation closer to real-world fidelity. The proposed future directions — advanced control strategies such as MPC and LQR, hardware-in-the-loop simulation, vision-based precision landing, and multi-vehicle coordination — represent natural next steps in a development pathway that could ultimately lead to physical hardware implementation and deployment.

The broader significance of this work lies in its validation of a rigorous simulation-first methodology for autonomous UAV development. Building and validating a high-fidelity simulation before committing resources to hardware fabrication reduces risk, accelerates the iteration cycle, and provides a deeper understanding of system behavior that informs better design decisions. In a domain as technically demanding and physically consequential as autonomous aerial delivery, this approach represents not just best engineering practice, but a fundamental prerequisite for safe and reliable operation.

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APPENDIX A: SIMULATION PARAMETERS SUMMARY

This appendix compiles all numerical parameters used in the quadcopter delivery simulation for reproducibility and transparency. Any future simulation built upon this work should use these values as the baseline.

Category	Parameter	Value	Unit
Physical	Total mass (drone + package)	1.5	kg
Physical	Drone mass only (post-release)	1.2	kg
Physical	Package mass	0.3	kg
Physical	Arm length (centre to motor)	0.225	m
Propeller	Diameter	10	inch (0.254 m)
Propeller	Thrust coefficient C_t	1.28×10^{-7}	N/RPM ²
Propeller	Torque coefficient C_ϕ	7.3×10^{-9}	N·m/RPM ²
Motor	First-order time constant τ_m	0.02	s
Motor	Maximum speed	8000	RPM
Motor	KV rating (reference)	950	RPM/V
Inertia	I^{xx} (roll)	0.0221	kg·m ²
Inertia	I^{yy} (pitch)	0.0221	kg·m ²
Inertia	I^{zz} (yaw)	0.0407	kg·m ²
Environment	Air density ρ	1.225	kg/m ³
Environment	Gravity g	9.81	m/s ²
Environment	Body drag coefficient C_s	0.02	N·s/m
Package	Package drag coefficient	0.01	N·s/m

Control	Position loop rate	50	Hz
Control	Attitude loop rate	500	Hz
Control	Release tolerance ϵ	0.3	m
Simulation	Total mission time	35	s
Simulation	Solver type	ODE45	—
Simulation	Maximum solver step size	0.001	s

Table A.1: Complete Simulation Parameter Set

APPENDIX B: MOTOR MIXER ALLOCATION MATRIX

The motor mixer translates the four aggregate control inputs $[U_1, \tau\phi, \tau\theta, \tau\psi]$ into the four individual motor thrust commands $[T_1, T_2, T_3, T_4]$. In matrix form:

$$[T_1, T_2, T_3, T_4]^T = M \cdot [U_1, \tau\phi, \tau\theta, \tau\psi]^T$$

The allocation matrix M is:

	U_1	$\tau\phi$	$\tau\theta$	$\tau\psi$
T_1	1/4	0	$-1/(2L)$	$+1/(4K)$
T_2	1/4	$+1/(2L)$	0	$-1/(4K)$
T_3	1/4	0	$+1/(2L)$	$+1/(4K)$
T_4	1/4	$-1/(2L)$	0	$-1/(4K)$

Table B.1: Motor Mixer Allocation Matrix M

Substituting $L = 0.225$ m and $K = C_\phi/C_t = (7.3 \times 10^{-9}) / (1.28 \times 10^{-7}) \approx 0.057$ gives the numerical mixing matrix used in the simulation. The inverse operation — computing aggregate control inputs from measured motor thrusts — uses the same matrix structure with the signs reversed.

APPENDIX C: TRAJECTORY COEFFICIENT DERIVATION

This appendix documents the procedure for computing the polynomial trajectory coefficients for a single flight segment. Consider a segment from waypoint A at time t_0 to waypoint B at time t^0 . The desired trajectory $p(t)$ is expressed as:

$$p(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5$$

Six boundary conditions are imposed at the segment endpoints. For a stop-and-go trajectory (zero velocity and acceleration at each waypoint):

$$\text{At } t = t_0: \quad p(t_0) = p_A, \quad \dot{p}(t_0) = 0, \quad \ddot{p}(t_0) = 0$$

$$\text{At } t = t^0: \quad p(t^0) = p_B, \quad \dot{p}(t^0) = 0, \quad \ddot{p}(t^0) = 0$$

Evaluating the polynomial and its derivatives at these boundary conditions yields the 6×6 linear system:

$$\begin{bmatrix} 1, & t_0, & t_0^2, & t_0^3, & t_0^4, & t_0^5; & 0, & 1, & 2t_0, & 3t_0^2, & 4t_0^3, & 5t_0^4; & 0, & 0, & 2, & 6t_0, & 12t_0^2, & 20t_0^3; & 1, & t^0, & t^{0^2}, & t^{0^3}, & t^{0^4}, & t^{0^5}; & 0, & 1, & 2t^0, & 3t^{0^2}, & 4t^{0^3}, & 5t^{0^4}; & 0, & 0, & 2, & 6t^0, & 12t^{0^2}, & 20t^{0^3} \end{bmatrix} \cdot [a_0; a_1; a_2; a_3; a_4; a_5] = [p_A; 0; 0; p_B; 0; 0]$$

This system is solved numerically in MATLAB at simulation initialization using the backslash operator (left matrix division), which exploits the structure of the Vandermonde-type matrix for efficient solution. The coefficients are stored in memory and evaluated at each simulation time step to generate the real-time reference signals.

For segments in which the yaw angle must change smoothly, the same polynomial interpolation procedure is applied independently to the yaw angle ψ , treating it as a separate one-dimensional trajectory with its own set of six boundary conditions. The resulting yaw trajectory is guaranteed continuous in angle, angular rate, and angular acceleration, ensuring that the yaw controller is never subjected to step changes in its reference input.

--- END OF REPORT ---

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